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Benchmark solutions

Entropic lattice Boltzmann simulation of thermal convective turbulence

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ABSTRACT

We present a comprehensive study of thermal convective turbulence using a two-population entropic multi-relaxation lattice Boltzmann model. Both the canonical non-rotating and rotating Rayleigh–Bénard problems are addressed by the lattice Boltzmann simulation in the turbulent regime, ranging from high Rayleigh numbers convection in the former to geostrophic turbulence in the latter. Special attention is payed to evaluating the lattice Boltzmann method as an alternative for coarse-grid simulation in the rapidly rotating Rayleigh–Bénard regime. Results obtained for the Nusselt number scaling and flow structure are compared to direct numerical simulation and predictions of the reduced non-hydrostatic quasi-geostrophic equations. It is found that, even under a severe under-resolution of Ekman layer, the present lattice Boltzmann model is capable of capturing various regimes of thermal convective turbulence in a quantitative fashion. This study demonstrates that lattice Boltzmann method can serve an alternative to conventional turbulence modeling approaches.

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1. Introduction

Thermal convective turbulence is ubiquitous in Nature and technological applications. In the geophysical and astrophysical context, convective motion driven by temperature stratification under the effect of gravity takes place in stars and planets [1–4], as well as in atmospheric and oceanic circulation [5–8]. Buoyancy-driven flows also occur in biological systems in the thermal-bio-convection phenomena [9,10]. In engineering and technological applications, thermal convection plays important role in process technology [11], heat exchangers and chip cooling [12], as well as in the quality of indoor circulation [13]. Most of the above examples combine thermal convection with other relevant physical phenomena such as rotation of the frame of reference, radiative and conjugate heat transfer, phase change, compressibility or magneto-hydrodynamic effects, or chemical reactions [14].

A paradigm for these situations is the Rayleigh–Bénard convection (RBC) [15,16] when the reference frame is at rest, or the rotating Rayleigh–Bénard convection (RRBC) in the case of rotating reference frame [17,18]. RBC plays a prominent role in the development of stability theory in hydrodynamics [19,20] as well as in pattern formation and spatial-temporal chaos [21,22]. Special attention was devoted for long to the non-rotating RBC because of its relevance to planetary systems, providing the limit behavior when convection overcomes rotational effects. In particular, convective heat transfer in RBC gives the upper bound on that of the rotating systems [18].

In RBC, a fluid layer of height H is heated from below, cooled from above, and is under the influence of buoyancy forces. The non-dimensional groups characterizing the system are the Rayleigh number Ra, representing the ratio between thermally induced buoyancy and the viscosity and thermal diffusivity forces, the Prandtl number Pr, representing the ratio between viscous and thermal diffusion, and the aspect ratio A of the geometry:

$$\mathrm{Ra} = \frac{g\lambda \Delta T H^3}{\nu\alpha}, \quad \mathrm{Pr} = \frac{\nu}{\alpha}, \quad A = \frac{H}{L}. \tag{1.1}$$

Here g is gravitational acceleration, λ is the thermal expansion coefficient, ΔT is the temperature difference between the bottom and top boundaries of the fluid layer, H is the distance between the boundaries, ν is the kinematic viscosity, α is the thermal diffusivity and L the horizontal characteristic size of the domain. When the Rayleigh number exceeds the critical value Ra_c, the system becomes unstable to convective motion. For the canonical Rayleigh–Bénard problem without rotation, with no-slip boundaries, the critical Rayleigh number is $\mathrm{Ra}_c = 1708$ [23].

The key response of the system to the imposed Rayleigh number is the heat flux h from bottom to top. The dimensionless

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heat flux,

$$\mathrm{Nu} = \frac{hH}{\alpha \Delta T}, \tag{1.2}$$

is the Nusselt number representing the ratio between the actual heat flux and its pure conductive part. With the Oberbeck–Boussinesq approximation [24,25], the Nusselt number for incompressible flow becomes

$$\mathrm{Nu} = \frac{\langle u_z T \rangle_A - \alpha \partial_z \langle T \rangle_A}{\alpha \Delta T / H}, \tag{1.3}$$

where $\langle \cdot \rangle_A$ denotes averaging over any horizontal plane and over the time. Nusselt number is a particularly important quantity in RBC since it provides a global characterization of the strength of convection. Much work has been done in order to estimate scaling laws for the turbulent heat transport, $\mathrm{Nu} = \mathrm{Nu}(\mathrm{Ra}, \mathrm{Pr}, A)$. Two main theories were verified by experiments: the classical scaling law of Malkus [26], $\mathrm{Nu} \sim (\mathrm{Ra}/\mathrm{Ra}_c)^{1/3}$, which mainly holds for $\mathrm{Ra} \gtrsim 10^{10}$ and another one, more appropriate for moderate regimes ($\mathrm{Ra} \lesssim 10^{10}$), with $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$ [27,28]. Other scaling laws were proposed for different Rayleigh and Prandtl number regimes [15,29,30]. The unifying theory by Grossmann and Lohse [31,32] provides a synthesis of various scaling regimes in a wide range of parameters.

With rotational effects included, new modes of convection may develop depending on the strength of rotation. Importance of viscosity against Coriolis forces is characterized by the Ekman number E,

$$\mathrm{E} = \frac{\nu}{2\Omega H^2}, \tag{1.4}$$

where Ω is the angular rotation rate. Rotation tends to suppress or delay the onset of convection [33]. The critical Rayleigh number for the onset of convection is not constant anymore but instead grows with the rotation rate as

$$\mathrm{Ra}_c = c_2 \mathrm{E}^{-4/3}, \tag{1.5}$$

where $c_2 \simeq 8.696$ as $\mathrm{E} \to 0$ [19]. Moreover, the Ekman number influences the turbulent heat transfer so that the parameter space becomes even wider with respect to RBC, i.e. $\mathrm{Nu} = \mathrm{Nu}(\mathrm{Ra}, \mathrm{E}, \mathrm{Pr}, A)$. By assuming that Malkus marginal boundary layer arguments [19] hold also for rotating systems, [34] predicted that rotating convective heat transfer scales as $\mathrm{Nu} \sim (\mathrm{Ra}/\mathrm{Ra}_c)^3$, and that this steep, cubic scaling holds from the onset of convection up to the Rayleigh number Ra_T at which the thermal boundary layer becomes nested within the Ekman layer, i.e. $\mathrm{Ra}_T \sim \mathrm{E}^{-3/2}$ [18]. Above Ra_T, the scaling $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$ sets in, in agreement with the non-rotating RBC.

RBC was studied extensively using both experiments and simulations. The goal is not only to explore the response of the system in a range of parameters but rather to find an asymptotic regime in the high-Rayleigh-number convection. With recent experiments, it was possible to reach $\mathrm{Ra} \simeq 10^{14} - 10^{17}$ [35–38] while direct numerical simulations (DNS) started exploring the transition regime $\mathrm{Ra} \simeq 10^{12}$ where the boundary layer becomes turbulent [14,39]. While a reasonably wide parameter range was studied for non-rotating Rayleigh–Béneard convection, it is different for the rotating RBC [40,41], see [42] and references therein [43–45]. Both experiments and DNS face difficulties in the turbulent but rotationally constrained regime [46]. Experiments can reach high levels of turbulence relatively easily but encounter problems in ensuring dominance of Coriolis forces in the force balance. On the other hand, DNS suffers from a large range of spatial and temporal scales that need to be resolved [46]. Contemporary experiments and DNS for RRBC are restricted to $\mathrm{E} \gtrsim 10^{-8}$, $\mathrm{Ra} \lesssim 10^{13}$ and $\mathrm{E} \gtrsim 10^{-7}$, $\mathrm{Ra} \lesssim 10^{12}$, respectively, according to [47].

Alternative modeling and simulation approaches were developed for both RBC and RRBC. For the non-rotating Rayleigh–Bénard convection, Large Eddy Simulation (LES) strategies were used as preferable path to reaching very high Rayleigh numbers [48–50]. However, in order to capture the near-wall stresses and heat transfer, grid resolution requirements become similar to DNS in the near-wall region. Even higher values of Rayleigh number up to $\mathrm{Ra} \simeq 2 \cdot 10^{16}$, were addressed by the transient RANS simulations [51,52]. However, due to the empirical nature of modeling and parameter tuning, prediction strength reduces [50]. Concerning RRBC, a system of reduced equations valid in the limit $(\mathrm{Ro}_c, \mathrm{E}) \to 0$, where $\mathrm{Ro}_c = \mathrm{E}\sqrt{\mathrm{Ra}/\mathrm{Pr}}$ is the convective Rossby number, was developed by Julien et al. [53,54], non-hydrostatic quasi-geostrophic equations (NH-QGE). Simulations using NH-QGE allowed to discover a rich dynamical behavior in the low Ekman - high Rayleigh number regimes. However, NH-QGE are limited to the stress-free boundary conditions; thus, while the behavior described by this reduced set of equations in the bulk of the flow is well reproduced, the heat transfer in the near-wall regions may be affected as compared to the no-slip boundary [46]. Recently, the NH-QGE system was extended to include the influence the effect of Ekman pumping by including parametric boundary conditions (composite NH-QGE (CNH-QGE), [41,55]), and their implications on the large-scale dynamics are currently a research topic [40,41].

More recently, the lattice Boltzmann method (LBM) was gaining attention as an alternative for fluid dynamics [56,57]. In particular, the entropic LBM formulation demonstrates interesting applications ranging from turbulent flows in complex moving geometries [58], transitional flows [59], multiphase [60], compressible flows [61] and magnetohydrodynamics [62], to mention some recent examples. LBM is a recast of the fluid dynamics equations into a kinetic framework for discretized distribution function (populations) $f_i = f_i(\boldsymbol{x}, t)$ at position $\boldsymbol{x}$ and time t, associated with discrete velocities $\boldsymbol{v}_i$, $i = 1, \ldots, Q$, and designed to recover the Navier–Stokes equations in the hydrodynamic limit. Discrete velocities fit into a regular Cartesian lattice; this allows efficient "stream-along-links-and-equilibrate-at-nodes" computation algorithm. Extension of LBM to thermal flows was considered by many authors through introducing another set of populations g_i to recover the temperature dynamics in the Oberbeck-Boussinesq approximation (so-called two-population LBM; different versions were presented, e. g. by He et al. [63–66]). Many works reported two-dimensional RBC simulation as a matter of validation, or the vertical or other kind of natural convection [67–71]. Three-dimensional LBM simulations of RBC turbulence and related setups have been used to demonstrate, for the first time, the so-called ultimate [29] scaling $\mathrm{Nu} \sim \mathrm{Ra}^{1/2}$ in the turbulent bulk (without boundaries) [72]; to address and observe the Bolgiano-Obukhov scaling of structure functions [73,74], to detect anisotropies in the scaling exponents of velocity and temperature structure functions [75]; to study the effect of an imposed wind on the heat flux scaling [76]. Extensions to convection in particle-laden [77] and multiphase flows [78] were also addressed. Typical range of Rayleigh numbers addressed in these works is of the order of $\mathrm{Ra} \sim 10^7$. However, to-date, only a limited number of works can be found in the literature for the simulation of non-rotating RBC at high Rayleigh number, of the orders of the state-of-the-art DNS and, to the best of our knowledge, none for the rotating RBC.

In this paper, we present simulations of three-dimensional RBC and RRBC for a wide range of Rayleigh numbers, and for different rotation rates, in the range of state-of-the-art DNS. We demonstrate capabilities of the recently introduced two-population entropic multi-relaxation lattice Boltzmann model [65,66,79] for performing coarse-grid simulations. Differently from LES for RBC, coarse grids can be employed near the walls, while still retaining the accuracy necessary to capture first- and second-order statistics. Moreover, differently from NH-QGE, no-slip boundary conditions can be employed and, even under severe under-resolution of

Ekman layer, the main features of the flow are demonstrated. We thus demonstrate that the two-population entropic multirelaxation lattice Boltzmann model is a viable alternative to study thermal convective turbulence.

The paper is organized as follows: in Section 2, the two-population entropic multi-relaxation lattice Boltzmann model is presented, together with the implementation of gravity and Coriolis forces. On the technical side, the LBM model of Section 2 is a generalization of the previously introduced two-population models by Karlin et al. [65,66] where the entropic multi-relaxation principle is now extended to any Prandtl number. In Section 3, we present results for the non-rotating RBC for the Rayleigh number in the range $\mathrm{Ra} = 10^7 - 10^{11}$. Performance of LBM in the under-resolved simulation is evaluated through comparison of the flow structure with recent DNS study of [80]. In Section 4, we consider the rotating Rayleigh–Bénard problem for Ekman number in the mildly rotating range, $\mathrm{E} = 4.3 \cdot 10^{-3} - 5.6 \cdot 10^{-5}$, and compare results with the DNS of [17]. In Section 5, we study the rapidly rotating Rayleigh–Bénard convection (RRRBC) for $\mathrm{E} = 10^{-7}$ and Rayleigh number up to $\mathrm{Ra} = 0.5 \cdot 10^{12}$. The LBM simulation is shown to recover a series of different flow regimes of RRRBC, from cellular to geostrophic turbulence and we discuss emergence of large-scale vortex structures. Conclusions are drawn in Section 6.

2. Thermal entropic multi-relaxation lattice Boltzmann model

2.1. Kinetic equations

In the thermal two-population LBM, one considers two sets of kinetic equations for the populations f_i, responsible for the mass and momentum equations, and the populations g_i, responsible for the energy. The locally conserved fields are defined as follows:

$$\rho = \sum_{i=1}^{Q} f_i, \tag{2.1}$$

$$\rho \boldsymbol{u} = \sum_{i=1}^{Q} f_i \boldsymbol{v}_i, \tag{2.2}$$

$$2\rho E = D\rho T + \rho u^2 = \sum_{i=1}^{Q} g_i, \tag{2.3}$$

where ρ is the density, $\boldsymbol{u}$ is the flow velocity, E is the total energy, with D the space dimension ($D = 3$ below), and T is the temperature. LBM kinetic equations for f- and g-populations are compactly written [65,66],

$$\begin{aligned} f_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) &= f_i'(\boldsymbol{x}, t) = (1-\beta) f_i + \beta f_i^{mirr} \\ &\quad + 2(\beta - \beta_1)\left[f_i^* - f_i^{eq}\right], \end{aligned} \tag{2.4}$$

$$\begin{aligned} g_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) &= g_i'(\boldsymbol{x}, t) = g_i + 2\beta\left(g_i^{eq} - g_i\right) \\ &\quad + 2(\beta - \beta_1)\left[g_i^* - g_i^{eq}\right]. \end{aligned} \tag{2.5}$$

Here the left-hand side of both equations represents the advection and the right-hand side represents the post-collision populations f_i' and g_i'. Furthermore, f_i^{eq} and g_i^{eq} are the equilibrium populations, while f_i^* and g_i^* are the quasi-equilibrium populations. The choice of the quasi-equilibrium is dictated by the value of the Prandtl number, and two sets $\{f_i^*, g_i^*\}$ shall be specified below for two separate cases, $\mathrm{Pr} \leqslant 1$ and $\mathrm{Pr} > 1$. Relaxation parameters β and β_1 are related to kinematic viscosity ν and thermal diffusivity α. Finally, f_i^{mirr} is the over-relaxed mirror state of the f-populations. Note that, when the mirror state is specified as $f_i^{mirr} = 2f_i^{eq} - f_i$, Eq. (2.4) recover the conventional LBGK collision. Before identifying the mirror state of the entropic multi-relaxation LBM, we shall specify the equilibrium and quasi-equilibrium populations.

Table 1
Generating equilibria and quasi-equilibria using the form F_i (2.12).

Pr	F_i	M_0	M_α	$M_{\alpha\beta}$
any Pr	f_i^{eq}	ρ	ρu_α	$P_{\alpha\beta}^{eq}$
any Pr	g_i^{eq}	$2\rho E$	q_α^{eq}	$K_{\alpha\beta}^{eq}$
$\mathrm{Pr} \leqslant 1$	f_i^*	ρ	ρu_α	$P_{\alpha\beta}^{eq}$
$\mathrm{Pr} \leqslant 1$	g_i^*	$2\rho E$	$q_\alpha - 2u_\beta\left(P_{\alpha\beta} - P_{\alpha\beta}^{eq}\right)$	$K_{\alpha\beta}^{eq}$
$\mathrm{Pr} > 1$	f_i^*	ρ	ρu_α	$P_{\alpha\beta}$
$\mathrm{Pr} > 1$	g_i^*	$2\rho E$	$q_\alpha^{eq} + 2u_\beta\left(P_{\alpha\beta} - P_{\alpha\beta}^{eq}\right)$	$K_{\alpha\beta}^{eq}$

2.2. Equilibrium and quasi-equilibrium populations

Equilibrium populations f_i^{eq} are obtained by minimizing the entropy function H,

$$H(f) = \sum_{i=1}^{Q} f_i \ln\left(\frac{f_i}{W_i}\right), \tag{2.6}$$

subject to fixed mass and momentum, (2.1) and (2.2). We use the standard lattice with $Q = 27$ discrete velocities $\boldsymbol{v}_i = (v_{ix}, v_{iy}, v_{iz})$ constructed as tensor product of three copies of the fundamental one-dimensional set, $v_{i\alpha} \in \{-1, 0, 1\}$. For this D3Q27 lattice, the weights are: $W_i = \prod_\alpha^{D=3} W_{v_{i\alpha}}$, with $W_0 = 2/3$, and $W_{-1} = W_1 = 1/6$ [81]. Let us introduce the pressure tensor $P_{\alpha\beta}$ and the energy flux vector q_α,

$$P_{\alpha\beta} = \sum_{i=1}^{Q} v_{i\alpha} v_{i\beta} f_i, \tag{2.7}$$

$$q_\alpha = \sum_{i=1}^{Q} v_{i\alpha} g_i. \tag{2.8}$$

For low Mach numbers, the pressure tensor, the energy flux and the contracted fourth-order tensor $R_{\alpha\beta}^{eq}$ assume the following values at equilibrium [65]:

$$P_{\alpha\beta}^{eq} = \sum_{i=1}^{Q} v_{i\alpha} v_{i\beta} f_i^{eq} = \rho T_0 \delta_{\alpha\beta} + \rho u_\alpha u_\beta, \tag{2.9}$$

$$q_\alpha^{eq} = \sum_{i=1}^{Q} v_{i\alpha} g_i^{eq} = 2\rho E u_\alpha + 2\rho T_0 u_\alpha, \tag{2.10}$$

$$R_{\alpha\beta}^{eq} = \sum_{i=1}^{Q} v_{i\alpha} v_{i\beta} g_i^{eq} = 2\rho E\left(T_0\delta_{\alpha\beta} + u_\alpha u_\beta\right) + 2\rho T_0\left(T_0\delta_{\alpha\beta} + 2u_\alpha u_\beta\right), \tag{2.11}$$

where $T_0 = 1/3$ is the reference temperature of the $D3Q27$ lattice. For low Mach numbers, minimization of the entropy function can be treated using expansion to second order in the velocity $\boldsymbol{u}$. This allows to generate equilibrium and quasi-equilibrium populations from a quadratic form,

$$F_i = W_i\left(M_0 + \frac{M_\alpha v_{i\alpha}}{T_0} + \frac{\left(M_{\alpha\beta} - M_0 T_0 \delta_{\alpha\beta}\right)\left(v_{i\alpha} v_{i\beta} - T_0\delta_{\alpha\beta}\right)}{2T_0^2}\right), \tag{2.12}$$

where summation convention is assumed, and where the scalar M_0, the vector M_α and the tensor $M_{\alpha\beta}$ are specified in Table 1

Table 2
Simulations parameters and estimated length scales for Rayleigh–Bénard simulations. From left to right: Rayleigh number Ra, grid points in x- and y-directions N_x and N_y, grid points in the z-direction N_z, aspect ratio A, non-dimensional grid spacing $\Delta x/H = \Delta y/H = \Delta z/H$, non-dimensional dissipation scale η/H, non-dimensional Taylor micro-scale λ/H, non-dimensional thermal boundary layer δ_T/H, relative grid spacing with respect to dissipation scale $\Delta z/\eta$ and relative grid spacing with respect to thermal boundary layer $\Delta z/\delta_T$.

Ra	$N_x = N_y$	N_z	A	$\Delta x,\ \Delta y,\ \Delta z/H$	η/H	λ/H	δ_T/H	$\Delta z/\eta$	$\Delta z/\delta_T$
10^7	736	92	8	0.0109	0.0076	0.0834	0.0295	1.43	0.37
10^9	768	384	2	0.0026	0.0017	0.0406	0.0079	1.52	0.33
10^{11}	384	768	0.5	0.0013	0.0004	0.0271	0.0021	3.25	0.62

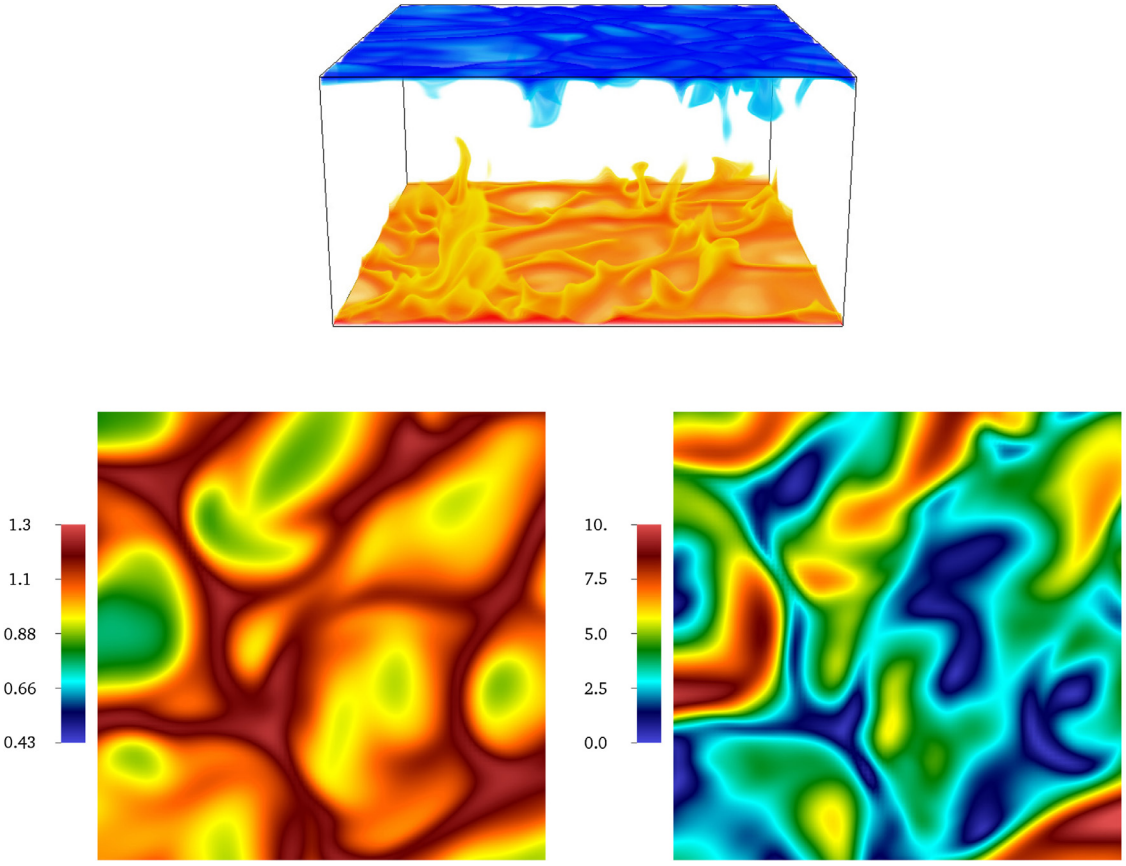


Fig. 1. Visualization of instantaneous flow field at $\mathrm{Ra} = 10^7$. Top: volume rendering of temperature T; Bottom left: slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω.

to obtain f_i^{eq}, g_i^{eq}, f_i^* and g_i^* upon substitution of pertinent combinations of vectors and tensors (2.7)–(2.11). Note that while the equilibrium populations are independent of the Prandtl number, the quasi-equilibria differ for $\mathrm{Pr} \leqslant 1$ and $\mathrm{Pr} > 1$. For $\mathrm{Pr} \leqslant 1$, the quasi-equilibrium f-populations are identical with the equilibrium, $f_i^* = f_i^{\mathrm{eq}}$, and the last term in the kinetic Eq. (2.4) vanishes. This model is analogous to Shakhov's S-model in the conventional kinetic theory [82] (the quasi-equilibrium g-population depends on the nonequilibrium energy flux (2.8)), and was first introduced by Karlin et al. [65] in the two-population LBM context. For $\mathrm{Pr} > 1$, the quasi-equilibrium f-population depends on the nonequilibrium pressure tensor (2.7), and this model is analogous to Levermore's model [83,84]. The need for two separate models in order to cover the entire range of Prandtl number was earlier highlighted by Ansumali et al. [85].

2.3. Entropic multi-relaxation formulation

Following [79], we now specify the mirror populations f_i^{mirr} in (2.4). To that end, let us recall the full set of natural moments of the D3Q27,

$$M_{pqr} = \sum_{i=1}^{Q} f_i v_{ix}^p v_{iy}^q v_{iz}^r, \quad p, q, r \in \{0, 1, 2\}. \tag{2.13}$$

Inverting the linear system (2.13), populations f_i are represented in terms of linear combination of moments M_{pqr} (expicit formulas can be found elsewhere, see e.g., [86]. or Appendix A of [58], and are not reproduced here). With this moment representation, populations are split in the following way,

$$f_i = k_i + s_i + h_i, \tag{2.14}$$

where k_i is the kinematic part which depends only on the locally conserved fields, $\rho = M_{000}$, $\rho u_x = M_{100}$, $\rho u_y = M_{010}$ and $\rho u_z = M_{001}$; the shear part s_i depends on the pressure tensor $P_{\alpha\beta}$ (2.7) ($P_{xy} = M_{110}$, $P_{yz} = M_{011}$, $P_{xz} = M_{101}$, $P_{xx} = M_{200}$, $P_{yy} = M_{020}$ and $P_{zz} = M_{002}$) while h_i lumps all the remaining higher-order moments. With this representation, the mirror-state f_i^{mirr} is written in the form:

$$f_i^{mirr} = k_i + \left[2s_i^{eq} - s_i\right] + \left[\gamma h_i + (1-\gamma) h_i^{eq}\right], \tag{2.15}$$

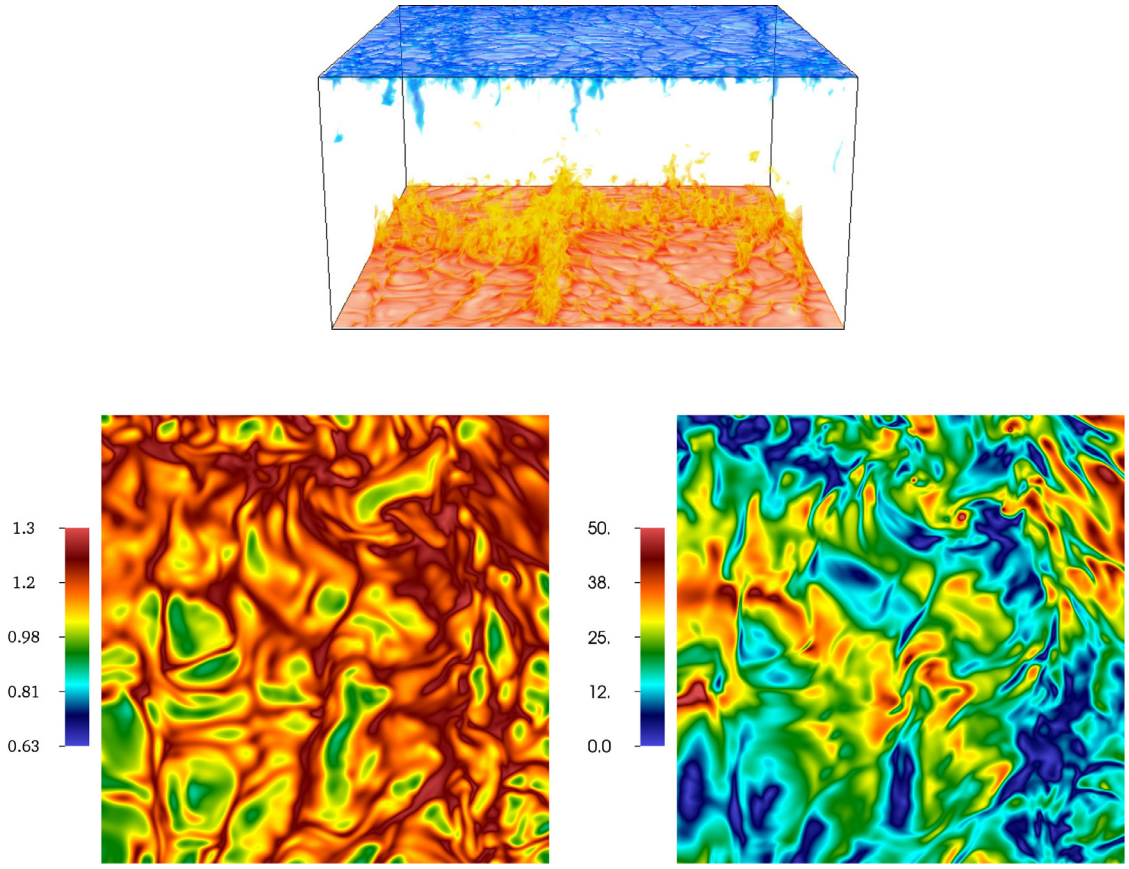


Fig. 2. Visualization of instantaneous flow field at $\mathrm{Ra} = 10^9$. Top: Volume rendering of temperature T; Bottom left: Slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: Slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω.

where the entropic stabilizer γ is computed by minimizing the entropy (2.6) in the post-collision state. It is sufficient to use first-order analytical approximation [79],

$$\gamma = \frac{1}{\beta_1} - \left(2 - \frac{1}{\beta_1}\right)\frac{\langle \Delta s | \Delta h\rangle}{\langle \Delta h | \Delta h\rangle} - \frac{2(\beta - \beta_1)}{\beta_1}\frac{\langle \Delta f^* | \Delta h\rangle}{\langle \Delta h | \Delta h\rangle}, \qquad (2.16)$$

where deviations from equilibrium are $\Delta s_i = s_i - s_i^{eq}$, $\Delta h_i = h_i - h_i^{eq}$ and $\Delta f_i^* = f_i^* - f_i^{eq}$, while $\langle \cdot | \cdot \rangle$ is the entropic scalar product, $\langle X | Y\rangle = \sum_{i=1}^{Q} X_i Y_i / f_i^{eq}$. For $\mathrm{Pr} \leqslant 1$, the last term in (2.16) vanishes because $f_i^* = f_i^{eq}$; in this case (2.16) reduces to the formula derived by Karlin et al. [79]. However, for $\mathrm{Pr} > 1$ the quasi-equilibrium populations f_i^* differ from the equilibrium (cf. Table 1); hence, (2.16) extends the entropic stabilizer to the entire range of Prandtl number. With this remark, the kinetic equations (2.4) and (2.5) are fully specified. Further details concerning the entropic multi-relaxation formulation were discussed by Karlin et al. [58,79,86], and particularly for the two-population thermal LBM see [66,87].

2.4. Buoyancy and coriolis forces

Forces are included via the exact difference method [88]. The post-collision f-populations in the kinetic equation (2.4) are modified according to:

$$f_i' \to f_i' + \left[f_i^{eq}(\rho, \boldsymbol{u} + \delta\boldsymbol{u}) - f_i^{eq}(\rho, \boldsymbol{u})\right], \qquad (2.17)$$

where $\delta\boldsymbol{u} = (\boldsymbol{F}/\rho)\delta t$, is the velocity increment due to the force $\boldsymbol{F}$, with the time step $\delta t = 1$. Below, two types of forces act on the system depending on the case of interest. For the non-rotating Rayleigh–Bénard convection, only the buoyancy force is included for which the Oberbeck-Boussinesq approximation is employed,

Table 3
Nusselt number comparison to scaling laws. Scaling 1 (valid for $\mathrm{Ra} \leqslant 10^{10}$): $\mathrm{Nu} = 0.16\ \mathrm{Ra}^{0.284}$ [18]; Scaling 2 (valid for $\mathrm{Ra} > 10^{10}$): $\mathrm{Nu} = 0.075\ \mathrm{Ra}^{0.322}$ [18].

Ra	Present	Scaling 1	Scaling 2
10^7	15.35	15.56	13.46
10^9	62.40	57.56	59.3
10^{11}	236.50	212.87	261.25

while in the case of rotation also the Coriolis force is applied,

$$\{F_x, F_y, F_z\} = \{-2\rho u_y \Omega, \quad 2\rho u_x \Omega, \quad \rho g \lambda (T - T_m)\}, \qquad (2.18)$$

where $T_m = T_0 + 0.5\Delta T$ is the mean temperature between hot and cold walls and where we set $\Omega = 0$ for the non-rotating RBC.

2.5. Hydrodynamic limit

Taken together, the two-population entropic multi-relaxation lattice Boltzmann model described in Sections 2.1–2.4 recovers hydrodynamic equations as follows [65]:

$$\partial_t \rho + \nabla \cdot (\rho \boldsymbol{u}) = 0, \qquad (2.19)$$

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\frac{1}{\rho}\nabla(T_0 \rho) - \frac{1}{\rho}\nabla \cdot \boldsymbol{\Pi} + \frac{1}{\rho}\boldsymbol{F}, \qquad (2.20)$$

$$\partial_t T + \boldsymbol{u} \cdot \nabla T = -\frac{T_0}{c_v}\nabla \cdot \boldsymbol{u} - \frac{1}{\rho c_v}(\nabla \boldsymbol{u}) : \boldsymbol{\Pi} + \frac{1}{\rho}\nabla \cdot (\rho \alpha \nabla T), \qquad (2.21)$$

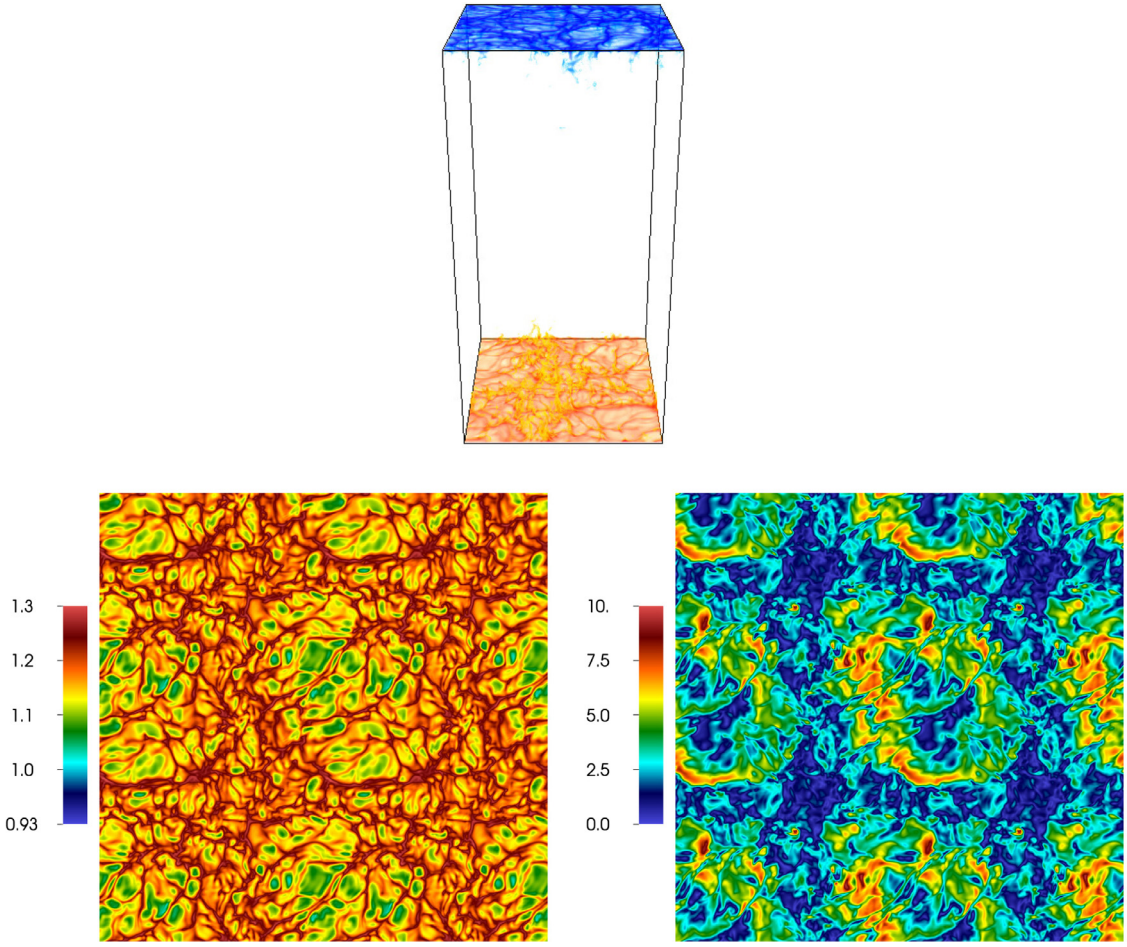


Fig. 3. Visualization of instantaneous flow field at $\mathrm{Ra} = 10^{11}$. Top: volume rendering of the temperature T; Bottom left: slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω.

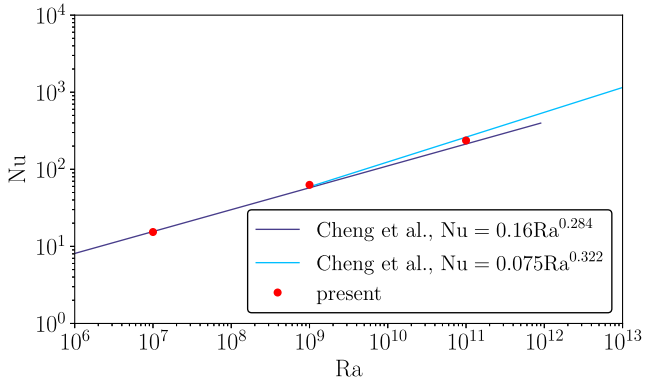


Fig. 4. Mean Nusselt number for the different Rayleigh numbers considered in RBC.

where $c_v = D/2$ is the specific heat of a D-dimensional ideal gas and $\boldsymbol{\Pi} = -\rho\nu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\dagger)$ is the viscous stress tensor. Kinematic viscosity ν and thermal diffusivity α are expressed through relaxation parameters β and β_1 as follows, depending on the Prandtl number

$$\nu = \left(\frac{1}{2\beta} - \frac{1}{2}\right)T_0, \qquad \alpha = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)T_0, \qquad \text{if } \mathrm{Pr} \leqslant 1, \tag{2.22}$$

$$\nu = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)T_0, \qquad \alpha = \left(\frac{1}{2\beta} - \frac{1}{2}\right)T_0, \qquad \text{if } \mathrm{Pr} > 1, \tag{2.23}$$

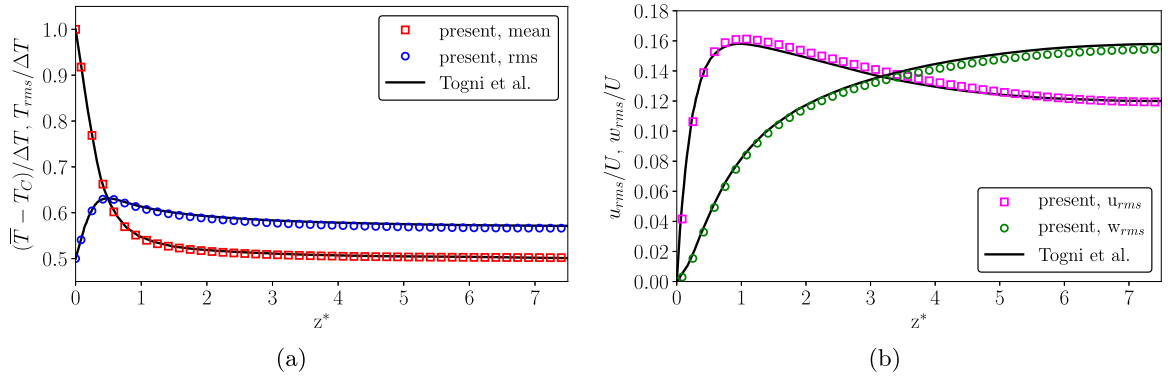
so that the Prandtl number $\mathrm{Pr} = \nu/\alpha$ reads

$$\mathrm{Pr} = \frac{(1-\beta)\beta_1}{(1-\beta_1)\beta}, \text{ if } \mathrm{Pr} \leqslant 1; \quad \mathrm{Pr} = \frac{(1-\beta_1)\beta}{(1-\beta)\beta_1}, \text{ if } \mathrm{Pr} > 1. \tag{2.24}$$

Note that contributions of the first and of the second term on the right hand side of the temperature Eq. (2.21) (compression work and viscous dissipation, respectively) are negligible in the Rayleigh–Bénard set-up under consideration. Hence, the two-population lattice Boltzmann model maintains the standard quasi-incompressible Oberbeck-Boussinesq approximation.

2.6. Simulations set-up

Simulations were performed in a three-dimensional rectangular domain with the dimensions $[L_x \times L_y \times L_z] = [HA \times HA \times H]$, where A is the aspect ratio. Cold T_C and hot T_H temperatures are fixed on the top and bottom walls, respectively, while the no-slip boundary condition for the velocity is applied at both walls. The wall boundary conditions were realized through Grad's approximation [66]. In the x- and y-directions, periodic boundary conditions are used in all the simulations in order to approximate an infinite horizontal extent ($A \to \infty$). This approximation is valid when the largest characteristic horizontal length-scales appearing naturally in the solution are smaller than the horizontal extent HA. The aspect ratio A was adapted for each simulation in order to comply with this assumption at a reasonable computational cost. The characteristic free-fall velocity, $U = \sqrt{g\lambda\Delta TH}$ was maintained below $U < 0.06$. All statistical quantities were collected after the initial transient of the



(a) (b)

Fig. 5. Temperature and velocity field statistics. (a): Mean and root mean square temperature along vertical direction; (b): Root mean square x- and z-components of velocity along vertical direction. Symbol: present LBM; Line: DNS of [80].

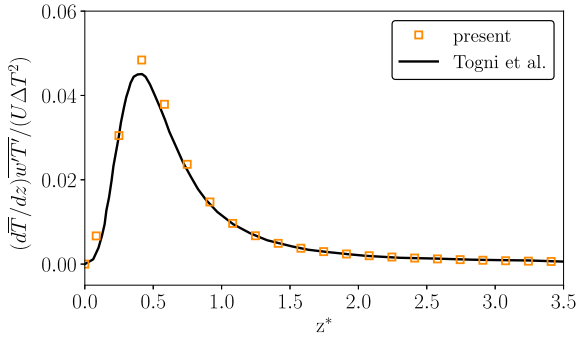


Fig. 6. Production of root mean square temperature along vertical direction. Symbol: present LBM; Line: DNS of [80].

simulation, and until statistical convergence was reached. Statistics were collected every $t_s = 5t_c$, where $t_c = H/U$ is the large eddy turnover time, in order to avoid correlation of the fields.

3. Non-rotating Rayleigh-Bénard convection

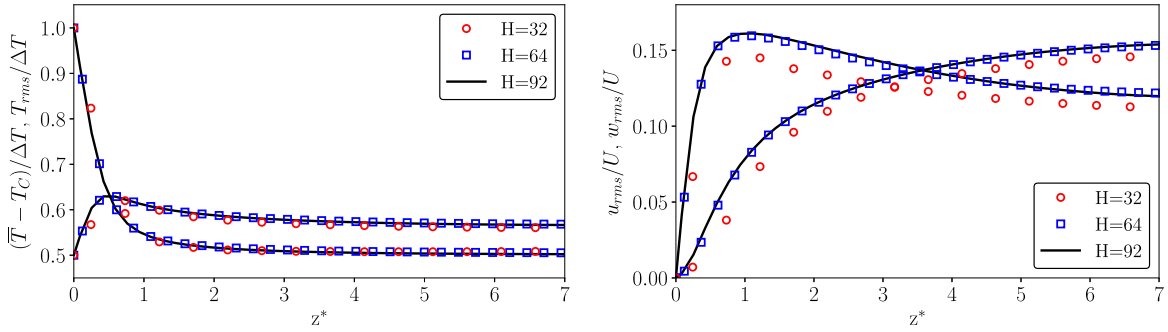
3.1. Parameter range

Before presenting results for the non-rotating RBC, a discussion of various length scales involved in the simulations is in order. As a general rule, in the DNS simulations, the minimal grid size is bounded by two key scales, i.e. dissipation scale $\eta = (\nu^3/\epsilon)^{1/4}$, where ϵ is the dissipation of turbulent kinetic energy [89,90], and the viscous and thermal boundary layers thickness, δ and δ_T, respectively. In DNS, it is common to resolve the layers with at least six grid points [50]. Specifically for RBC, [89] estimates the dissipation scale as $\eta/H = \sqrt{\mathrm{Pr}}/[(\mathrm{Nu}-1)\mathrm{Ra}]^{1/4}$. For LES simulations, the requirement on grid resolution is often expressed in terms of mean grid spacing, which is usually set to be at least half of the Taylor micro-scale λ, according to [50], $\lambda/H = \sqrt{15}u^*/[(\mathrm{Nu}-1)\mathrm{Ra}]^{1/2}$, where u^* is also estimated, and is given by Peng et al. [50,89]: $u^* = \sqrt{0.0027\mathrm{Ra}^{1.04} + 0.0312\mathrm{Ra}^{0.92}}$. Moreover, LESs typically require resolution similar to that of the DNS in the near-wall region in order to resolve the thermal boundary layer [50], $\delta_T/H = 0.5/\mathrm{Nu}$. All simulations of this section were at Prandtl number $\mathrm{Pr} = 0.7$.

Table 2 summarizes the cases simulated in this part of our study, together with the parameters and grid sizes as well as various length scales estimates. Note that, for the estimate of Nusselt number, correlations $\mathrm{Nu} = 0.16\ \mathrm{Ra}^{0.284}$ and $\mathrm{Nu} = 0.075\ \mathrm{Ra}^{0.322}$ were used for $\mathrm{Ra} \le 10^{10}$ and $\mathrm{Ra} > 10^{10}$, respectively [18]. Table 2 reveals that the dissipation scale is somewhat under-resolved for $\mathrm{Ra} = 10^7$ and $\mathrm{Ra} = 10^9$, and even more for $\mathrm{Ra} = 10^{11}$. However, previous studies of sub-grid features of KBC model for various non-thermal flows [58,59,87] give us some confidence about capturing low-order statistical properties of the flow at this degree of under-resolution. On the other hand, the thermal boundary layer is resolved with about three grid points for $\mathrm{Ra} = 10^7$ and $\mathrm{Ra} = 10^9$, and with at most two points for $\mathrm{Ra} = 10^{11}$. This renders the present simulations particularly challenging since both DNS and LES use a much more resolved grid in near-wall regions. We shall continue discussion of resolution requirements for the thermal KBC model in Sections 3.4 and 3.5.

3.2. Flow morphology

In Figs. 1–3, instantaneous volume rendering of temperature is shown for the three cases of Rayleigh number, together with



(a) (b)

Fig. 7. Temperature and velocity field statistics for the different girds $H = 32$, $H = 64$ and $H = 92$. (a): Mean and root mean square temperature distribution along vertical direction; (b): Root mean square x- and z-components of velocity distribution along the vertical direction.

Table 4
Nusselt number evaluation with two different methods, Eqs. (3.1) and (3.2), and for three different grids $H = 32$, $H = 64$ and $H = 92$. Reported is also the relative error ϵ with respect to the reference value $\mathrm{Nu}_{\mathrm{DNS}} = 15.59$ of [80].

H	32	64	92
$\mathrm{Nu} = 1 + \frac{\sqrt{\mathrm{RaPr}}\langle wT\rangle_V}{U\Delta T}$, Eq. (3.1)	12.33	14.93	15.35
Relative error ϵ [%]	20.9	4.2	1.5
$\mathrm{Nu} = -\frac{\partial\overline{T}}{\partial z}\frac{H}{\Delta T}$, Eq. (3.2)	10.89	14.63	15.48
Relative error ϵ [%]	30.15	6.16	0.7

a slice of temperature and normalized vorticity $\omega = |\omega|H/U$, near the bottom wall, at the half-height of the thermal boundary layer $z = 0.5\delta_T$. For rendering, red toward orange/yellow colors represent hot temperatures while blue toward light blue colors represent cold temperatures.

For all three cases, thermal plumes are clearly visible as localized portions of fluid with a temperature contrast to the background. With the increase of Rayleigh number, thermal plumes become smaller and less coherent. This is also reflected in the temperature slices near the bottom wall, where thermal plumes can be clearly distinguished by the dark red color. Hot and cold plumes detaching from the bottom and the top walls, respectively, are responsible for the large amount of heat transport. As can be seen clearly in Figs. 1–3, thermal plumes have a mushroom-like shape, while close to the walls they become sheet-like. With the help of the slices of temperature, one can see that sheet-like plumes create a thin network surrounding thermal cells while mushroom-like plumes are emitted from the intersection spots, as shown in the volume renderings. For lower Rayleigh number, plumes are of the order of domain size and are distributed more or less homogeneously over the domain. When the Rayleigh number is increased, thermal plumes tend to organize at the borders of convection cells which become more coherent and visible. This is attributed to a large-scale circulation ("wind") developing progressively stronger with increasing Rayleigh number [91–93]. A decrease of plumes size is also clearly visible on the slice of vorticity, Fig. 3.

3.3. Nusselt number scaling

Different ways of computing the Nusselt number exist; resorting to the conductive and convective heat transfer definitions, one can write, following [89]:

$$\mathrm{Nu} = 1 + \frac{\sqrt{\mathrm{RaPr}}\langle wT\rangle_V}{U\Delta T}, \tag{3.1}$$

where $\langle\cdot\rangle_V$ represent the average over the entire volume. Another approach computes the Nusselt number directly from the heat conduction at the walls,

$$\mathrm{Nu} = -\frac{\partial\overline{T}}{\partial z}\bigg|_w \frac{H}{\Delta T}, \tag{3.2}$$

where mean temperature gradient is evaluated by a semi-second-order finite difference scheme.

Results for the Nusselt number are reported in Table 3 where Eq. (3.1) was used for evaluation in all three cases. Good comparison is observed with respect to scaling laws as reported by Cheng et al. [18], in agreement with the 2/7 scaling for $\mathrm{Ra} < 10^{10}$ [27,28,94] and the steeper 1/3 scaling for $\mathrm{Ra} > 10^{10}$ [26] when the boundary layer becomes turbulent. Note that, even using Eq. (3.1) leads in an under-prediction of the Nusselt number when the under resolution is severe, as it is in the case $\mathrm{Ra} = 10^{11}$. This is be attributed to the fact that at $\mathrm{Ra} = 10^{11}$ only one to two grid points reside in the thermal boundary layer. Nevertheless, a fair estimate for the Nusselt number is observed in all cases.

3.4. Flow field analysis at $\mathrm{Ra} = 10^7$

The case $\mathrm{Ra} = 10^7$ is particularly interesting here since grid requirements in both LES [50] and DNS [80] can be compared to LBM. Note that both LES and DNS use non-uniform grids with more resolution at the walls, unlike the LBM which operates on a uniform Cartesian grid. As compared to the DNS of [80], the LBM grid is 2.35 times finer in the x- and y-directions. However, in the z-direction from bulk to wall, the LBM grid is 1.31 to 84.12 times coarser than DNS. Refinement in the horizontal directions happens simply because LBM uses a Cartesian grid with constant spacing, and by itself is not a resolution requirement. Similarly, the LES grid of [50] is up to 5.19 times finer in the near-wall region in the vertical direction. To summarize, while LBM operates on a somewhat finer grid in the horizontal direction, it features a much coarser grid in near-wall regions, also with respect to LES. In fact, while LES can use coarser grid in the bulk, it still requires DNS-like resolution near the walls [50].

The case $\mathrm{Ra} = 10^7$ is taken here as a reference for which a more detailed study is performed, thanks to DNS data of [80]. Mean temperature $\overline{T}$ and root mean square (rms) temperature T_{rms} are presented in Fig. 5a as a function of reduced vertical coordinate (normal to the bottom wall), $z^* = z\mathrm{Nu}/H$. While the thermal boundary layer is resolved with only three points in the LBM simulation ($\Delta z/\delta_T = 0.37$, see Table 2), both first- and second-order statistics compare well with DNS. The mean and rms temperatures demonstrate typical RBC behavior: after $\overline{T}$ drops through the thermal boundary layer, a state of perfect mixing $T_m = (T_H + T_C)/2$ is achieved and continues until the opposite wall is approached; T_{rms} goes through a maximum in the thermal boundary layer where plumes emission is most frequent thus enhancing mixing with

Table 5
Simulations parameters and grid sizes for rotating Rayleigh–Bénard convection. From left to right: compensated Rayleigh number $\widetilde{\mathrm{Ra}}$, Rayleigh number Ra, Ekman number E, number of grid points in x- and y-direction N_x and N_y, number of grid points in z-direction N_z, non-dimensional grid spacing $\Delta x/H = \Delta y/H = \Delta z/H$ and relative grid spacing with respect to thermal boundary layer $\Delta z/\delta_T$.

Case	$\widetilde{\mathrm{Ra}}$	Ra	E	$N_x = N_y$	N_z	Δx, Δy, $\Delta z/H$	$\Delta z/\delta_T$
RRBC1	21.4	$3.09 \cdot 10^4$	$4.27 \cdot 10^{-3}$	128	64	0.0156	0.05
RRBC2	28.1	$7.03 \cdot 10^4$	$2.83 \cdot 10^{-3}$	128	64	0.0156	0.08
RRBC3	44.6	$2.81 \cdot 10^5$	$1.41 \cdot 10^{-3}$	128	64	0.0156	0.15
RRBC4	57.2	$5.91 \cdot 10^5$	$9.76 \cdot 10^{-4}$	128	64	0.0156	0.20
RRBC5	92.9	$2.53 \cdot 10^6$	$4.72 \cdot 10^{-4}$	256	128	0.0078	0.16
RRBC6	138.7	$8.44 \cdot 10^6$	$2.58 \cdot 10^{-4}$	256	128	0.0078	0.23
RRBC7	207.2	$2.81 \cdot 10^7$	$1.42 \cdot 10^{-4}$	384	192	0.0052	0.21
RRBC8	326.5	$1.10 \cdot 10^8$	$7.15 \cdot 10^{-5}$	384	192	0.0052	0.31

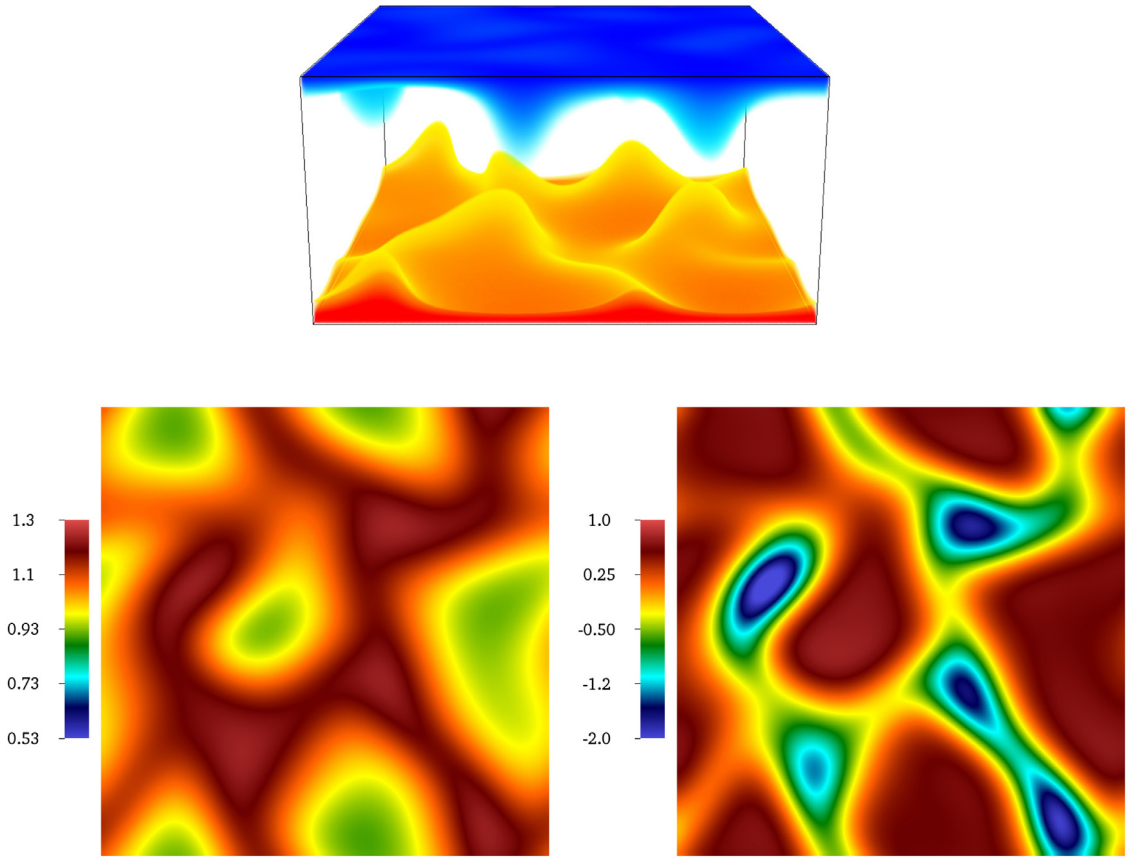


Fig. 8. Visualization of instantaneous flow field at $\mathrm{Ra} = 3.09 \cdot 10^4$. Top: Volume rendering of temperature T; Bottom left: Slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: Slice at $z = 0.5\delta_T$ of instantaneous vertical vorticity ω_z.

the bulk flow. Fig. 5b shows the rms of x- and z-components of velocity, u_{rms} and w_{rms}, respectively. LBM results compare well with DNS. Vertical w_{rms} and horizontal u_{rms} behave differently. According to [80], the vertical rms velocity, produced by buoyancy, reaches its maximum in the bulk and is transported toward the boundary by pressure mechanisms. In the boundary layer, w_{rms} is redistributed to the horizontal rms velocity components by pressure-strain rate making u_{rms} to reach maximum. The horizontal rms velocity components are both transported to the bulk and redistributed in the boundary layer where both rms components are dissipated. Finally, in Fig. 6 we compare a more subtle quantity, the production of the rms of temperature, $\varsigma = -2\overline{w'T'}(d\overline{T}/dz)$, which includes correlation of vertical velocity and temperature $\overline{w'T'}$ (turbulent kinetic energy production), where $T' = T - \overline{T}$, and $d\overline{T}/dz$ is the mean vertical temperature gradient. While a small deviation in the peak value is noted, agreement between LBM and DNS is reasonable also in this case. Summarizing, in spite of a much coarser grid in the near-wall region, LBM compares well with DNS results for a range of first- and second-order statistics.

3.5. Sub-grid features and grid requirement

In order to further probe into the sub-grid features of the thermal LBM, we run the $\mathrm{Ra} = 10^7$ case on different grids. In Fig. 7, the mean and rms temperatures, and rms x- and z-velocity are reported for a coarse $H = 32$ and a finer $H = 64$ grids in comparison with the results for $H = 92$ reported in previous section. For a very coarse grid $H = 32$, the boundary layer in the mean temperature is overestimated, while the rms temperature and both rms velocities are underpredicted. However, the main physical mechanism of the flows is still qualitatively captured even with such a severely coarse grid. For the finer grid $H = 64$, the statistics already does not show any appreciable deviation from the correct values reported for a grid of $H = 92$, even if also in this case the under-resolution is quite significant.

Table 4 reports Nusselt number evaluated with both Eqs. (3.1) and (3.2). For the coarsest grid $H = 32$, Nusselt number is under-predicted by at least $\epsilon = 20.9\%$, with both ways of evaluation, as the result of over-estimated thermal boundary layer. However, with the grid $H = 64$, the error drops below $\epsilon = 5\%$ when evaluated with the mean convective heat flux (3.1), while direct computation of the gradient in Eq. (3.2) results in a somewhat larger error attributed to remaining slight over-estimation of the thermal boundary layer at this still coarse resolution. Indeed, for $H = 64$, Kolmogorov scale is underestimated by a factor $\Delta z/\eta = 2.06$ while the thermal layer is resolved with only about two grid points: $\delta_T/\Delta z = 1.89$.

Summarizing, for the test-case $\mathrm{Ra} = 10^7$ the required minimum resolution is achieved with Cartesian grid $H = 64$. Implicit sub-grid property of entropic LBMs has been known for some time, e. g., [58,59,87]. Here, it indicates that LBM can be useful for thermal convection simulation on relatively coarse grids. These estimations also explain the under-prediction of the Nusselt number of the case $\mathrm{Ra} = 10^{11}$ in Table 3. In this case, only less than two points are present within the theoretical size of the thermal boundary layer; such an under-resolution leads in fact to an over-estimation of the thermal boundary layer size, and consequently of an underestimation of the Nusselt number.

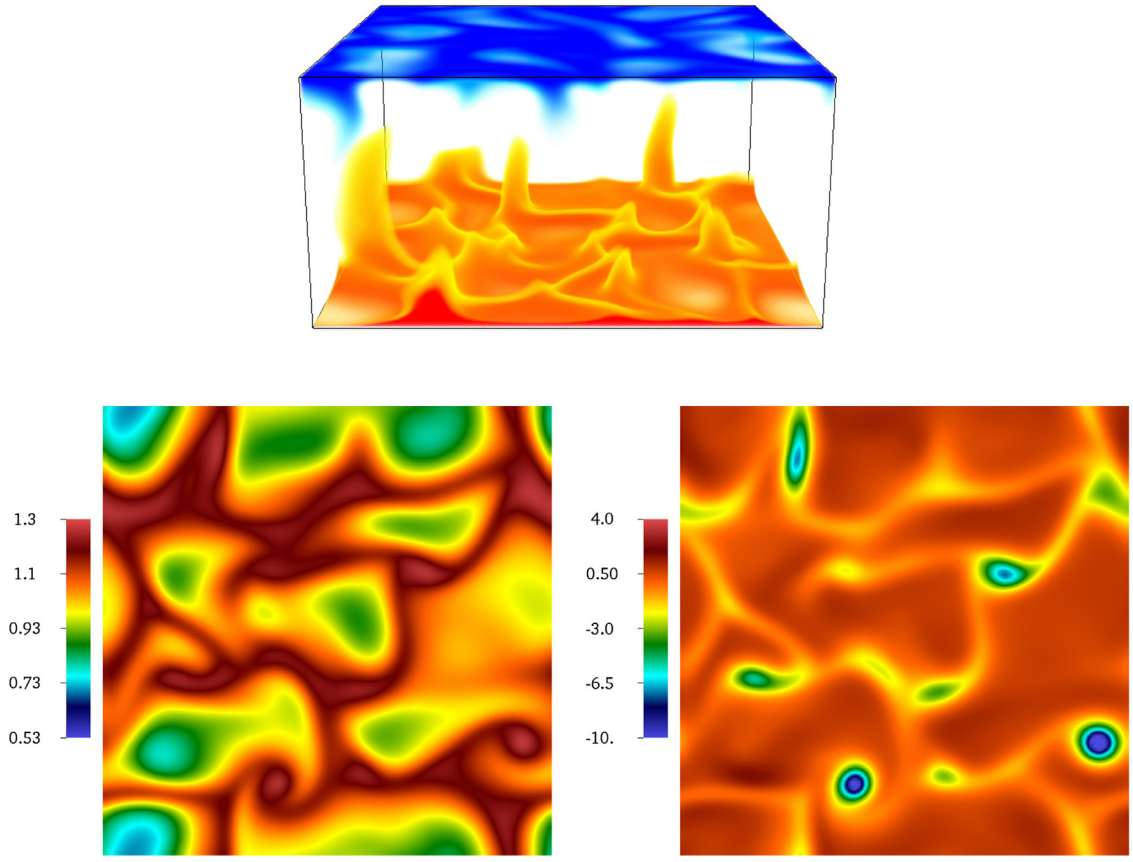


Fig. 9. Visualization of instantaneous flow field at $\mathrm{Ra} = 5.91 \cdot 10^6$. Top: Volume rendering of temperature T; Bottom left: Slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: Slice at $z = 0.5\delta_T$ of instantaneous vertical vorticity ω_z.

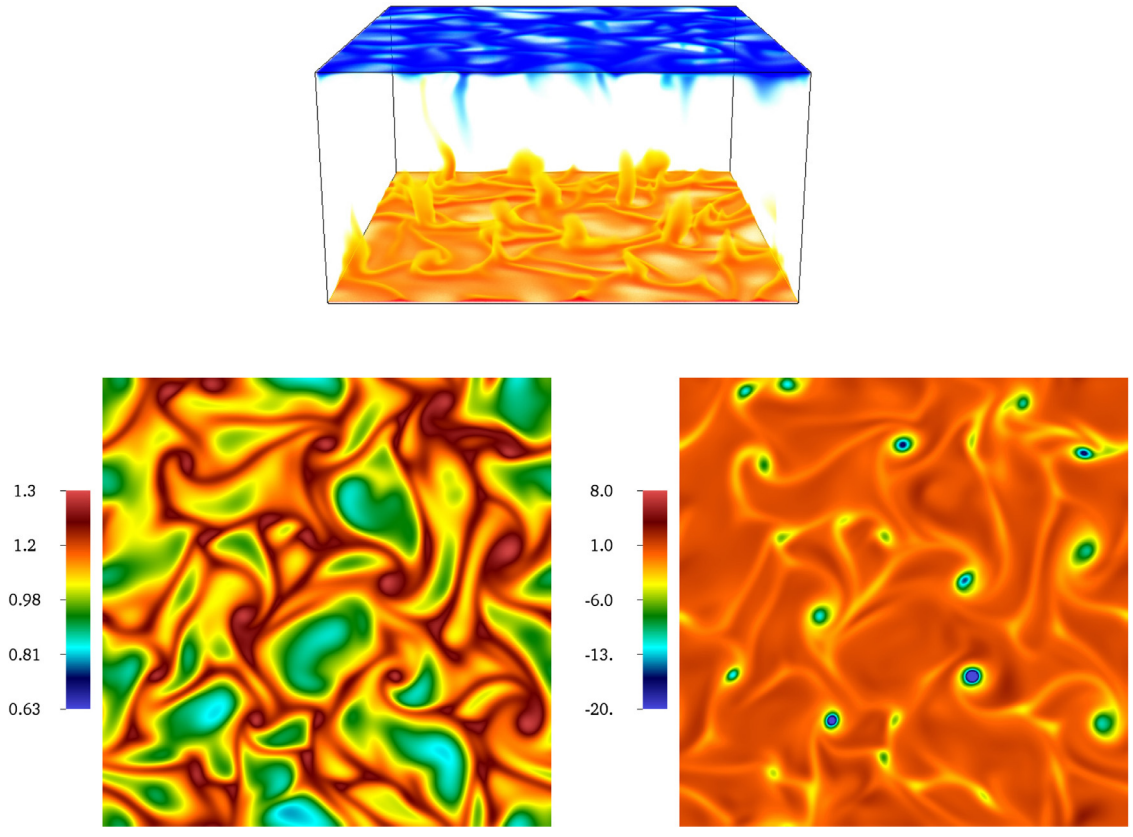


Fig. 10. Visualization of instantaneous flow field at $\mathrm{Ra} = 8.44 \cdot 10^6$. Top: Volume rendering of temperature T; Bottom left: Slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: Slice at $z = 0.5\delta_T$ of instantaneous vertical vorticity ω_z.

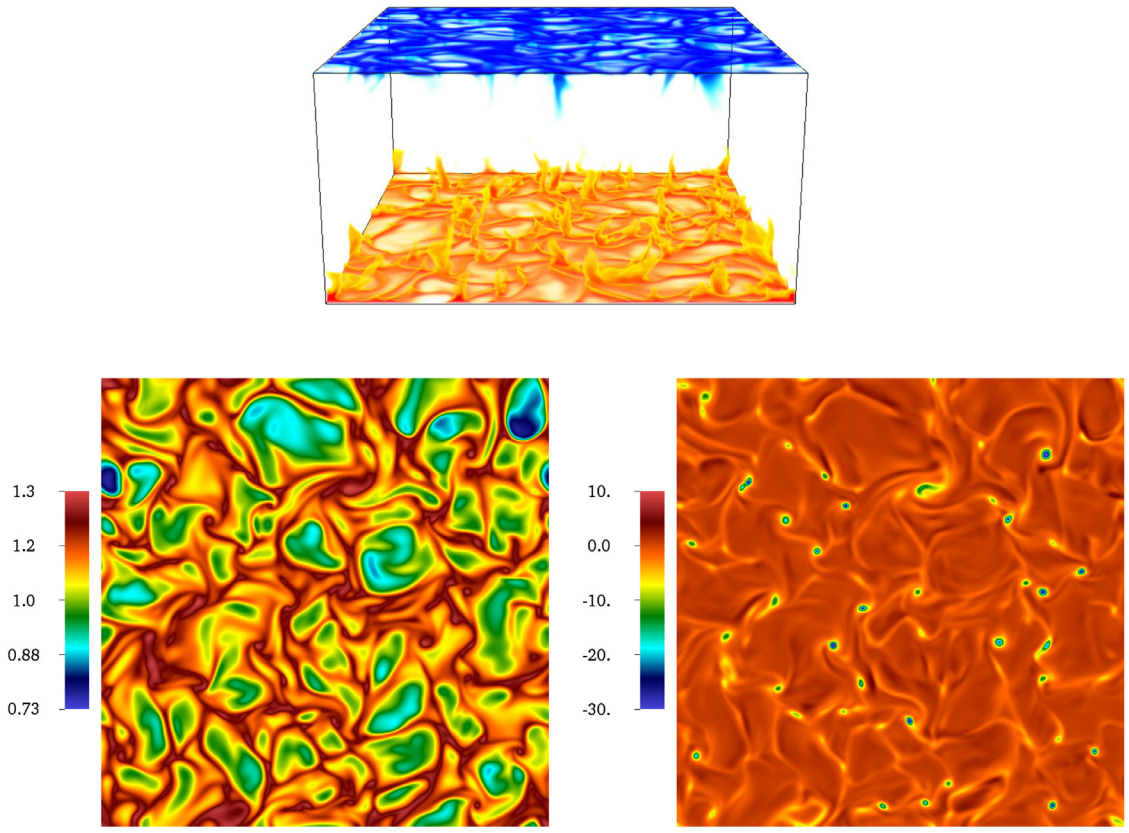


Fig. 11. Visualization of instantaneous flow field at $\mathrm{Ra} = 1.1 \cdot 10^8$. Top: Volume rendering of temperature T; Bottom left: Slice at $z = 0.5\delta_T$ of instantaneous temperature T; Bottom right: Slice at $z = 0.5\delta_T$ of instantaneous vertical vorticity ω_z.

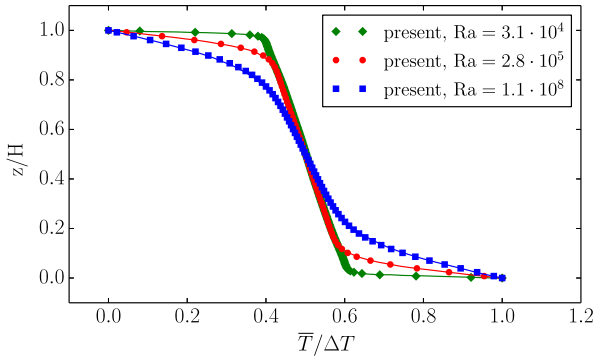


Fig. 12. Mean temperature profiles in RRBC. Three cases shown correspond to RRBC1 ($\mathrm{Ra} = 3.09 \cdot 10^4$), RRBC3 ($\mathrm{Ra} = 2.81 \cdot 10^5$) and RRBC8 ($\mathrm{Ra} = 1.10 \cdot 10^8$) of Table 5. Symbol: present LBM; Line: reference DNS of [17].

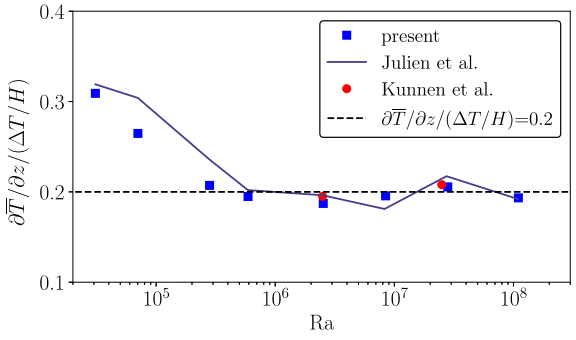


Fig. 13. Negative of mean temperature gradient at mid-layer as a function of Raleigh number. Squares: present LBM, cases RRBC1-RRBC8 of Table 5; Circles: reference work of [43]; Full line: reference DNS of [17]. Asymptotic value is indicated with dashed line.

4. Rotating Rayleigh–Bénard convection

4.1. Parameter range

We now proceed with the results for the rotating Rayleigh–Bénard convection (RRBC) where the viscous fluid layer is under the effect of rotation. This is reflected by the Ekman number (1.4) that decreases with increasing rotation rate. Owing for the estimate for the critical Rayleigh number (1.5), it is convenient to introduce the compensated Rayleigh number $\widetilde{\mathrm{Ra}} = \mathrm{Ra}\,\mathrm{E}^{4/3}$ as a measure of super-criticality [46]. In this section we consider weak to medium rotational strength, $\mathrm{E} = 4.3 \cdot 10^{-3} - 5.6 \cdot 10^{-5}$, for various Rayleigh numbers. Eight cases were considered, the respective simulation parameters are reported in Table 5. Resolution of the thermal boundary layer was maintained in a range from twenty down to three grid points, from RRBC1 to RRBC8. In accord with the analysis in the previous sections, this complies with the minimum required resolution. All simulations were done with no-slip boundary conditions, and will be compared with the DNS of [17]. Prandtl number is $\mathrm{Pr} = 1$ and convective Rossby number $\mathrm{Ro}_c = \mathrm{E}\sqrt{\mathrm{Ra}/\mathrm{Pr}}$ is $\mathrm{Ro}_c = 0.75$ in all simulations.

4.2. Flow morphology

Flow field is visualized exemplary for four cases of Table 5 in Figs. 8 (RRBC1)– 11 (RRBC8). Each case is demonstrated with instantaneous volume renderings of temperature and a slice of tem-

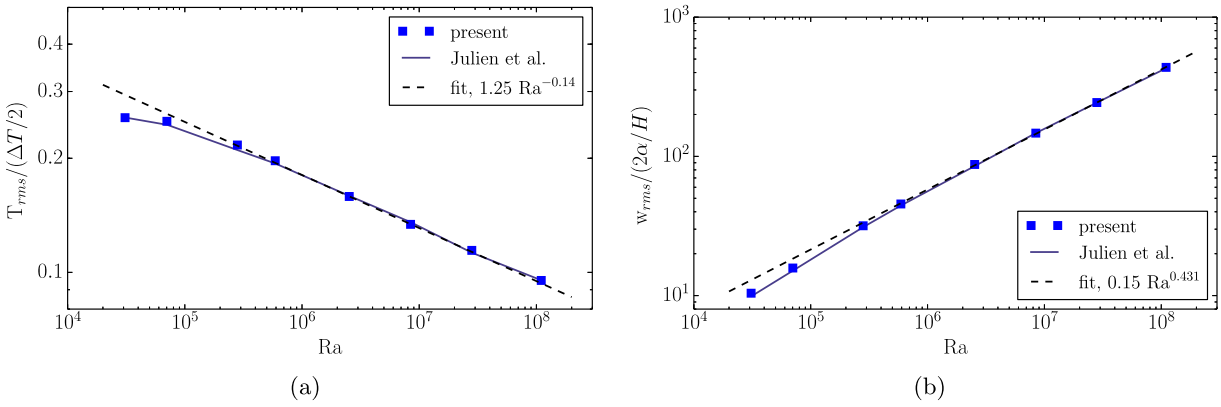


Fig. 14. Second-order statistics of RRBC on the mid-plane. (a) Reduced root mean square temperature T_{rms}. (b) Reduced root mean square vertical velocity w_{rms}. Symbol: present LBM; Full line: reference DNS [17]. Asymptotic scaling laws, $T_{rms} \sim \mathrm{Ra}^{-0.14}$ and $w_{rms} \sim \mathrm{Ra}^{0.431}$ [17] are indicated with dashed lines.

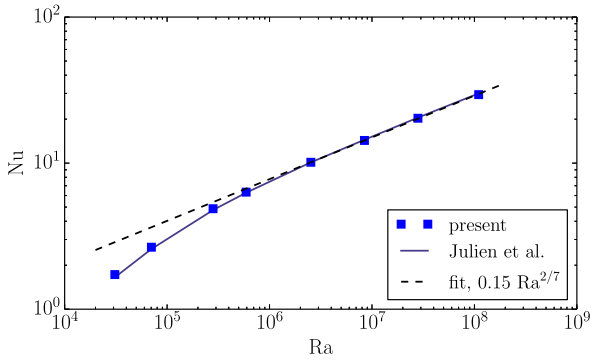


Fig. 15. Nusselt number dependence on Rayleigh number in RRBC. Symbol: present LBM; Full line: reference DNS [17]. Non-rotating RBC scaling, $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$ is shown with dashed line.

perature and of vertical vorticity ω_z near bottom wall, at the half-height of the thermal boundary layer $z = 0.5\delta_T$. The case RRBC1 in Fig. 8 for low $\mathrm{Ra} = 3.09 \cdot 10^4$ is characterized by convection-rolls pattern associated with Küppers-Lortz instability [17], clearly visible in both temperature and vertical vorticity. With Rayleigh number increasing, convection rolls give way to chaotic interactions between vertical vortices associated with convection cells [17]. This scenario is realized in the RRBC4 simulation shown in Fig. 9 for $\mathrm{Ra} = 5.91 \cdot 10^5$. Thermal plumes start to be distinguishable in the volume rendering by the mushroom-like shapes but also in the temperature slice by the dark red color. This state persists up to Rayleigh number of the order $\mathrm{Ra} \sim 6 \cdot 10^5$, above which thin thermal plumes spontaneously appear in the flow field. Thermal plumes prevail near the junctions of convection cells where cell boundaries merge with the thermal boundary layers attached to the hot and cold walls [17]. As the Rayleigh number is further increased above $\mathrm{Ra} > 2 \cdot 10^6$ the flow field becomes dominated by coherent plumes while the cellular convection pattern becomes progressively more difficult to discern. This states are realized with the RRBC6 and RRBC8 simulations shown in Fig. 10 for $\mathrm{Ra} = 8.44 \cdot 10^6$ and Fig. 11 for $\mathrm{Ra} = 1.1 \cdot 10^8$. It is interesting to notice that, unlike in the non-rotating thermal convection, the vertical vorticity associated with thermal plumes is always cyclonic (of the same sign with the externally applied rotation). This is reflected in the vertical vorticity slices of Figs. 9– 11, where vertical vorticity is negative (blue) in the zones associated with thermal plumes, also discernible as the intersection of the boundary thermal cells.

4.3. Flow field analysis

We proceed with quantitative comparison of flow field of RRBC with the DNS of [17]. Mean temperature profiles for different Rayleigh numbers are shown to match very well with the DNS in Fig. 12. Fig. 12 demonstrates important implication of rotation on thermal convection, that is finiteness of mean temperature gradient $\partial\overline{T}/\partial z$ in the bulk of fluid layer. This is in contrast to non-rotating RBC, where the mean temperature gradient in the bulk vanishes if Rayleigh number is high enough. The magnitude of mean temperature gradient as a function of Rayleigh number is shown in Fig. 13. Both the present LBM simulation and the reference DNS of [17] consistently confirm saturation of the mean temperature gradient at a value of about $\left|(\partial\overline{T}/\partial z)(H/\Delta T)\right| \approx 0.2$. Explanation for this saturation invokes RRBC-specific vortical flow associated with thermal plumes: Cyclonic character of vortices mentioned above induces their co-rotation (attraction and merging). Co-rotation and merging promotes mixing of the fluid along horizontal planes ("lateral mixing", [17]); this impedes penetration of buoyant elements at greater distances away from the boundaries causing stratification of the temperature and hence a finite mean temperature gradient.

Second-order statistics in the bulk compares well with reference DNS over a wide range of Rayleigh number, see Fig. 14. Note that the scaling found in [17] is reproduced for both rms temperature and vertical velocity, $T_{rms} \sim \mathrm{Ra}^{-0.14}$ and $w_{rms} \sim \mathrm{Ra}^{0.431}$, respectively. Furthermore, the scaling exponent are consistent with those of the non-rotating RBC, $-0.14 \approx -1/7$ for T_{rms} and $0.431 \approx 3/7$ for w_{rms}. This corresponds to the free-fall behavior, also observed in the non-rotating RBC and indicates that Rayleigh number is sufficiently high so that dynamics of thermal plumes in the bulk are no longer controlled by the boundary conditions [17]. However, this does not imply that turbulence in the bulk is independent of the rotation rate. Indeed, the prefactor for $T_{rms} \sim \mathrm{Ra}^{-0.14}$ and $w_{rms} \sim \mathrm{Ra}^{0.431}$ depends on the convective Rossby number $\mathrm{Ro}_c = \mathrm{E}\sqrt{\mathrm{Ra}/\mathrm{Pr}}$ and differs from that of the non-rotating RBC. In fact, turbulence dynamics inherits a dependence on Ro_c from the near-boundary motions, which in turn are strongly affected by rotation.

4.4. Nusselt number scaling

It is interesting to note that, in the high Rayleigh number regime, the Nusselt number can by estimated as $\mathrm{Nu} \sim (2T_{rms}/\Delta T)(w_{rms}H/2\alpha)$. When the above approximate scaling $T_{rms} \sim \mathrm{Ra}^{-1/7}$ and $w_{rms} \sim \mathrm{Ra}^{3/7}$ are used, this estimate in turn gives $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$, similar to the scaling in the non-rotating RBC. This is confirmed in Fig. 15, where Nusselt number computed in the

Table 6
Simulations parameters and grid sizes for rapidly rotating Rayleigh–Bénard convection. From left to right: compensated Rayleigh number $\widetilde{\mathrm{Ra}}$, Rayleigh number Ra, Ekman number E, number of grid points in x-, y- and z-directions N_x, N_y and N_z, aspect ratio A, non-dimensional grid spacing $\Delta x/H = \Delta y/H = \Delta z/H$, thermal boundary layer δ_T, Ekman layer δ_E, relative grid spacing with respect to the thermal boundary layer $\Delta z/\delta_T$ and relative grid spacing with respect to the Ekman layer $\Delta z/\delta_E$.

$\widetilde{\mathrm{Ra}}$	Ra	E	$N_x = N_y$	N_z	A	$\frac{\Delta x, \Delta y, \Delta z}{H}$	$\frac{\delta_T}{H}$	$\frac{\delta_E}{H}$	$\frac{\Delta z}{\delta_T}$	$\frac{\Delta z}{\delta_E}$
11	$2.36988 \cdot 10^{10}$	10^{-7}	256	1024	0.25	0.000977	0.2451	0.000316	0.00399	3.0895
30	$6.46330 \cdot 10^{10}$	10^{-7}	256	1024	0.25	0.000977	0.0267	0.000316	0.03659	3.0895
90	$1.93899 \cdot 10^{11}$	10^{-7}	256	1024	0.25	0.000977	0.0086	0.000316	0.11333	3.0895
240	$5.17064 \cdot 10^{11}$	10^{-7}	256	1024	0.25	0.000977	0.0033	0.000316	0.29642	3.0895

present simulations is compared with the reference DNS. Also for the Nusselt number, the present simulation is in good agreement with the reference, and the correct scaling is recovered.

Summarizing, also for the rotating Rayleigh–Bénard problem, compliance with the minimum resolution requirement for thermal layer delivers excellent comparison to the DNS in a range of statistical quantities. Note that estimated Ekman layer, $\delta_E \sim \mathrm{E}^{1/2}H$ implies $4.1 \lesssim \delta_E \lesssim 1.6$ for all cases of Table 5. In other words, at least one to two grid points are preserved in the Ekman layer by the KBC simulation in the mildly rotating case.

5. Rapidly rotating Rayleigh–Bénard convection

5.1. Parameter range

Finally, we address the rapidly rotating Rayleigh–Bénard convection (RRRBC). Additional difficulty arises in this regime because of a relatively low value of Ekman number, $\mathrm{E} = 10^{-7}$ considered herein. Small Ekman number implies a thin Ekman layer of order $\sim \mathrm{E}^{1/2}H$. This renders RRRBC particularly challenging for state-of-the-art DNS where Ekman layer typically needs to be resolved by at least ten grid points [18].

We consider here four cases, with parameters and length scales summarized in Table 6. Note that, the three lower Rayleigh number cases correspond to the parameter range considered in the recent DNS studies of [41,46] while the fourth case of $\widetilde{\mathrm{Ra}} = 290$ corresponds to $\mathrm{Ra} = 0.517064 \cdot 10^{12}$ and is about 2.5 times higher than what was reported in the DNS literature, to the best of our knowledge. Based on estimates of grid spacing relative to thermal and Ekman layers, the challenge is the under-resolution of the latter and not of the former. In view of grid resolution analysis of non-rotating RBC in Section 3.5, the minimum of three grid points for resolving the thermal boundary layer is maintained at even the highest Rayleigh number. However, unlike the mildly rotating regime of Section 4, the Ekman layer is not resolved. It is well known that Ekman pumping (generation of vertical velocity) related to thin Ekman layer greatly influences heat transport at the boundaries [55]. In that regards, the so-called thermal wind or middle layer which mediates the enhanced transport of temperature fluctuations and vorticity between the Ekman layer and the bulk ([46,55,95], see e. g. Fig. 2 in [55]) and is of the order $O(\mathrm{E}^{1/3}H) \sim 4.75$, appears to be resolved by about four to five points in present simulation. Thus, the low Ekman number regime scrutinizes implicit sub-grid features of LBM.

5.2. Flow morphology

Each of the four cases listed in Table 6 represents a particular RRRBC flow regime exemplary demonstrated in Fig. 16 by volume rendering of the temperature anomaly,

$$\theta(x, y, z) = T(x, y, z) - \overline{T}(z), \tag{5.1}$$

where $\overline{T}(z)$ is the mean temperature on the corresponding plane at a given height.

The case $\widetilde{\mathrm{Ra}} = 11$ in Fig. 16a represents the cellular regime [46] which closely resembles the single-mode, or single-wavenumber solutions [54,96], with temperature anomaly θ, vertical velocity w and vertical vorticity ω_z exhibiting strong vertical coherence over the entire layer depth. Cellular columns are characterized by horizontal length scale, $l_{\mathrm{col}} = 4.8\mathrm{E}^{1/3}H = 22.8$, according to [46], in agreement with the seminal work of [97] and which is close to the mean spacing observed in the present simulation, $l_{col} \sim 25.6$.

At greater forcing, i.e. by increasing Ra, and for $\mathrm{Pr} \leqslant 1$ (see the case $\widetilde{\mathrm{Ra}} = 30$ in Fig. 16b), plumes-like structures form; thermal structures still spans over part of the depth of the fluid layer, but the strong vertical coherence of the cellular regime is lost. We note in passing that, for $\mathrm{Pr} > 1$ an intermediate state between cellular regime and plumes regime, the convective Taylor column regime, would appear [46]; however, this is beyond the scope of present study. By further increasing the thermal forcing, as for the case $\widetilde{\mathrm{Ra}} = 90$ of Fig. 16c, geostrophic turbulence regime sets in: all of the system-scale coherence is lost, and the flow field becomes anisotropic, with only some degree of coherence along the vertical direction with respect to the horizontal directions observable [47]. Finally, when the compensated Rayleigh number is further increased to $\widetilde{\mathrm{Ra}} = 240$ (Fig. 16d), a large-scale circulation flow forms on top of geostrophic turbulence. All flow structures shown in Fig. 16 agree qualitatively with the DNS and reduced system simulations reported in the literature for the specified parameter range [46,54,96].

5.3. Nusselt number scaling

Nusselt number is compared with the DNS of [46] in Fig. 17. While for lower Rayleigh number regimes, $\widetilde{\mathrm{Ra}} = 11, 30, 90$, LBM results are plotted against DNS data directly, the case $\widetilde{\mathrm{Ra}} = 240$, Fig. 16d, was compared with a fit on the significant DNS results ($\widetilde{\mathrm{Ra}} > 30$) since we failed to find published data for this very high Rayleigh number RRRBC regime. For the three highest $\widetilde{\mathrm{Ra}}$ cases, $\widetilde{\mathrm{Ra}} = 30, 90, 240$, comparison is very good, and which demonstrates capability of LBM to capture the net heat transfer correctly even without fully resolving the Ekman layer. For the lowest $\widetilde{\mathrm{Ra}} = 11$, the Nusselt number is somewhat under-predicted which probably attributes to insufficient resolution. Indeed, the Rayleigh number in this case is only slightly above the critical, $\mathrm{Ra}/\mathrm{Ra}_c \simeq 1.27$, where $\mathrm{Ra}_c \simeq 8.6956 \cdot \mathrm{E}^{-4/3}$. Closeness to the stable regime makes this special case particularly sensitive to under-resolution.

5.4. Large scale circulation

Using free-slip boundary conditions, a number of recent DNS and reduced models simulations reported a large-scale circulation in RRRBC when the parameters are within the range considered above, $\mathrm{E} = 10^{-7}$, $\widetilde{\mathrm{Ra}} \simeq 90$, see [46,47,96,98,99]. However, with no-slip boundary conditions, the situation is different. In the recent study of [41], neither DNS nor the Ekman pumping-supplemented

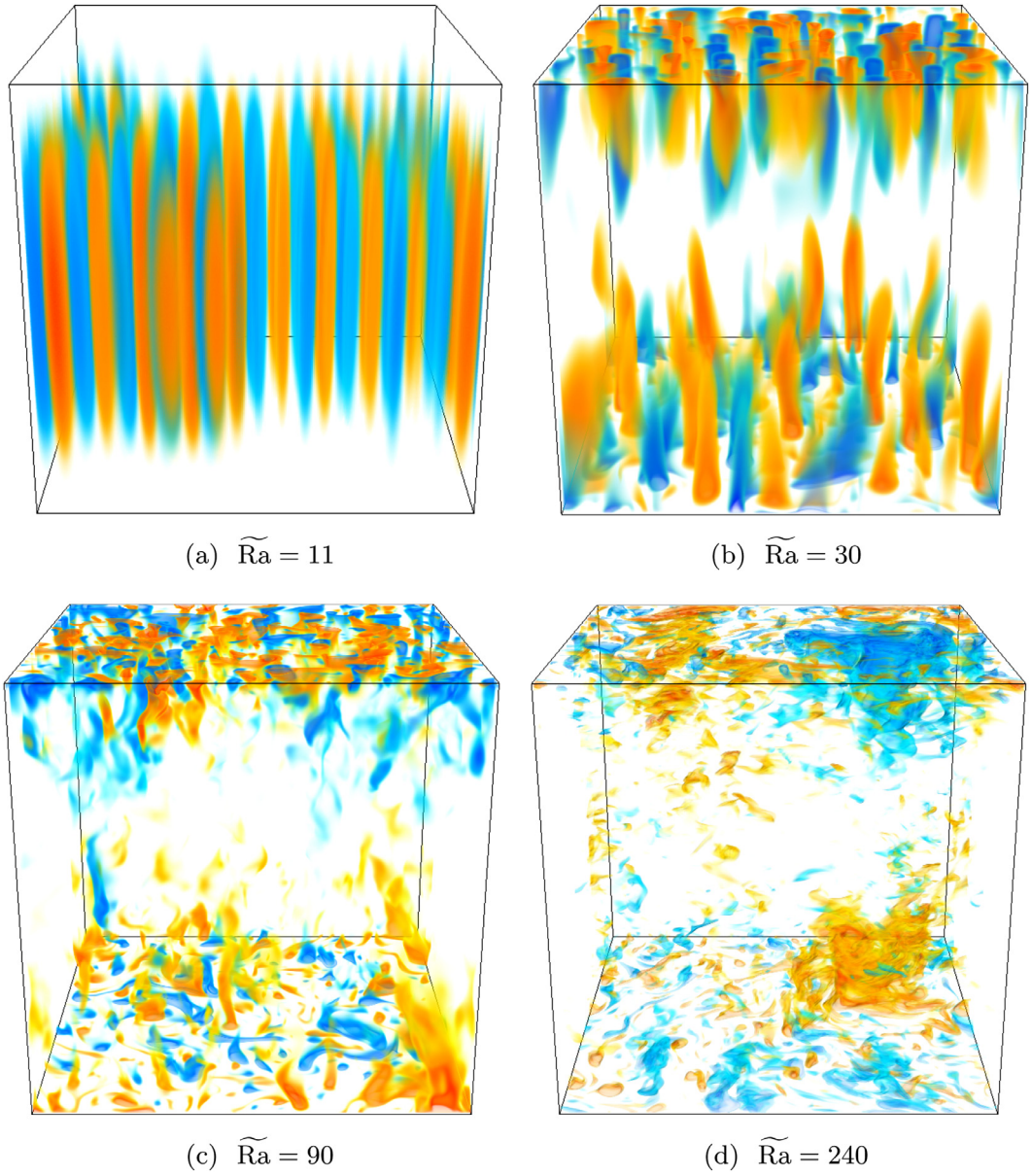


Fig. 16. Volume rendering of temperature anomaly θ for various flow regimes of rapidly rotating Rayleigh–Bénard convection at Ekman number $\mathrm{E} = 10^{-7}$ and Prandtl number $\mathrm{Pr} = 1$. (a) Cellular structure, $\widetilde{\mathrm{Ra}} = 11$; (b) Plumes structure, $\widetilde{\mathrm{Ra}} = 30$; (c) Geostrophic turbulence, $\widetilde{\mathrm{Ra}} = 90$; (d) Large scale circulation, $\widetilde{\mathrm{Ra}} = 240$.

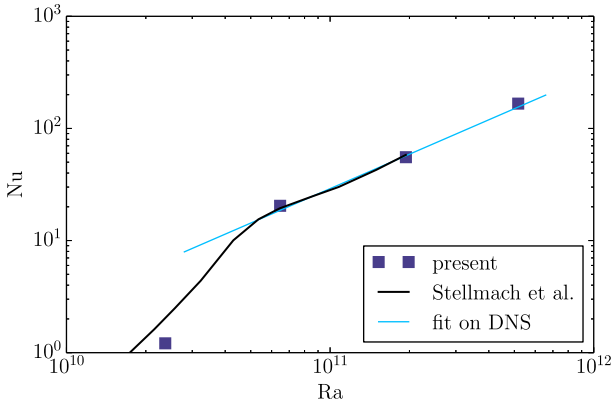


Fig. 17. Nusselt number as a function of Rayleigh number for the rapidly rotating Rayleigh–Bénard convection. Symbol: present LBM; Full line: reference DNS [46]. Thin line is a fit on DNS data for $\widetilde{\mathrm{Ra}} > 30$.

composite NH-QG model of [55] have found a persistent large-scale circulation at $\mathrm{E} = 10^{-7}$, $\widetilde{\mathrm{Ra}} \simeq 90$ and $\mathrm{Pr} = 1$. Moreover, DNS of [40] at a somewhat different parameter range argues that the no-slip boundary may even prevent formation of inverse cascade needed for spectral condensation. However, present simulation with no-slip boundary conditions demonstrates that persistent large-scale circulation structures do appear at higher Rayleigh numbers, as we have already pointed it out for the case $\mathrm{E} = 10^{-7}$, $\widetilde{\mathrm{Ra}} = 240$, see Fig. 16d corresponding to $\mathrm{Ra} = 0.517064 \cdot 10^{12}$. It is the goal of this section to quantify the large-scale circulation observed in this case.

To that end, it is convenient to introduce the vertically-averaged 2*D* horizontal flow, or the 2*D*-mode,

$$\overline{u}_z(x,y) = \frac{1}{H}\int_0^H u(x,y,z)dz, \quad \overline{v}_z(x,y) = \frac{1}{H}\int_0^H v(x,y,z)dz, \qquad (5.2)$$

where u and v are the velocity components in the x- and y- directions, respectively. Fast 3*D* fluctuations, or the 3*D*-mode, are de-

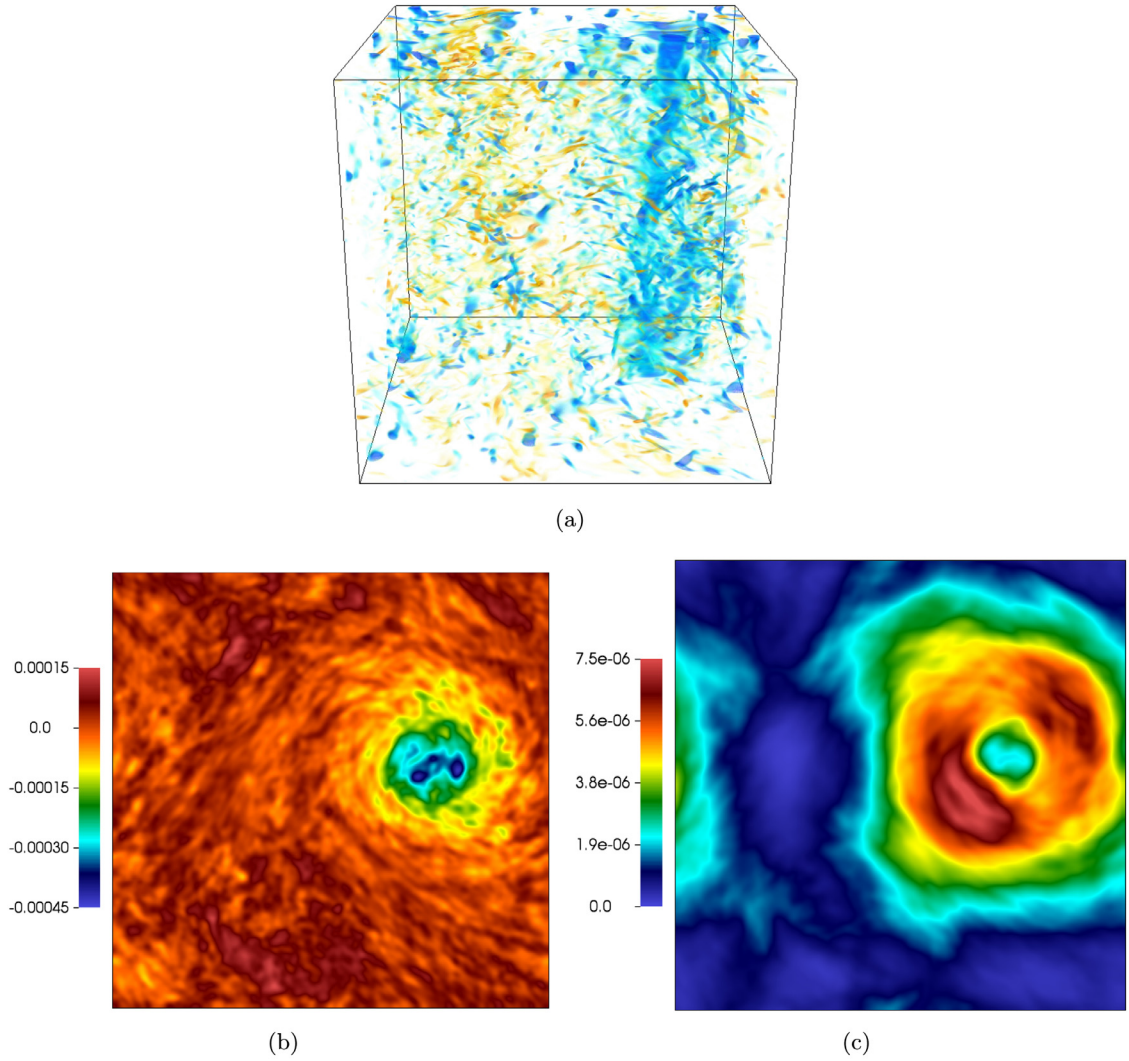


Fig. 18. Large scale vortex flow visualization for RRRBC at $\widetilde{\mathrm{Ra}} \simeq 240$. (a): Volume rendering of vertical vorticity ω_z; (b): Vertically averaged vertical vorticity ω_z; (c): Distribution of kinetic energy of the 2*D*-mode, $k_{2D} = (\overline{u}_z^2 + \overline{v}_z^2)/2$. .

fined as:

$$u'(x,y,z) = u(x,y,z) - \overline{u}_z(x,y), \tag{5.3}$$

$$v'(x,y,z) = v(x,y,z) - \overline{v}_z(x,y), \tag{5.4}$$

$$w'(x,y,z) = w(x,y,z). \tag{5.5}$$

The purely horizontal kinetic energy K_{2D} associated with the 2*D*-mode and the horizontal and vertical components of kinetic energy of the 3*D*-mode, K_{3D}^H and K_{3D}^V, respectively, are defined as follows:

$$K_{2D} = \frac{1}{2}\iint (\overline{u}_z^2 + \overline{v}_z^2)dxdy, \tag{5.6}$$

$$K_{3D}^H = \frac{1}{2}\iiint (u'^2 + v'^2)dxdydz, \tag{5.7}$$

$$K_{3D}^V = \frac{1}{2}\iiint w'^2 dxdydz. \tag{5.8}$$

Visualizations of the large-scale circulation flow for $\widetilde{\mathrm{Ra}} \simeq 240$ is presented by various means in Fig. 18. Volume rendering of the vertical vorticity ω_z in Fig. 18a shows formation of a large-scale vortex on the right, surrounded by smaller turbulent structures. Snapshots of the vertically averaged vertical vorticity, $(1/H)\int_0^H \omega_z dz$, and distribution of kinetic energy of the 2*D*-mode, $k_{2D} = (\overline{u}_z^2 + \overline{v}_z^2)/2$, are demonstrated in Fig. 18b and18c, respectively. Both Fig. 18b and 18c clearly show a large-scale vortex.

Table 7
Evolution of kinetic energy of 2*D*-mode and of 3*D* horizontal and vertical modes over six sequential time instances (time is in lattice units).

Time	$1\cdot10^6$	$2\cdot10^6$	$3\cdot10^6$	$4\cdot10^6$	$5\cdot10^6$	$6\cdot10^6$
$K_{2D}\times10^7$	1.0797	2.2362	3.4950	5.2812	14.204	17.450
$K_{3D}^H\times10^7$	1.4335	1.3084	1.1789	2.3976	3.6301	3.5805
$K_{3D}^V\times10^7$	1.2541	1.0852	1.0206	2.1239	3.1024	2.9864

History of the various components of kinetic energy is presented in Table 7. At the beginning of simulation, kinetic energy of the 2*D*-mode associated with the large-scale circulation is smaller compared to the 3*D*-modes. However, at later times, the 2*D*-mode becomes progressively more energetic, up to reaching the values of about five times larger than kinetic energy of each of the 3*D*-modes.

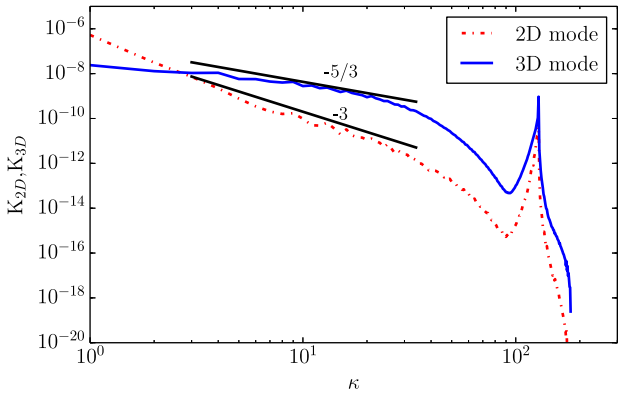


Fig. 19. Energy spectra of 2*D* (dashed line) and 3*D* (solid line) modes, $\kappa = \sqrt{\kappa_x^2 + \kappa_y^2}$ is the absolute value of the horizontal wave-vector. The 3*D*-mode displays inertial range $K_{3D} \sim \kappa^{-5/3}$ while the 2*D*-mode displays inertial range $K_{2D} \sim \kappa^{-3}$.

Finally, in order to provide further quantitative evidence of a large-scale circulation, Fig. 19 shows energy spectra of 2*D* and 3*D* modes (both horizontal and vertical components taken together for the latter). Down from the convective scale $\kappa_c \sim 4$, the inertial range of the 3*D*-mode is consistent with Kolmogorov's scaling, $K_{3D} \sim \kappa^{-5/3}$. On the other hand, energy spectrum of the 2*D*-mode demonstrates inertial range with the scaling $K_{2D} \sim \kappa^{-3}$, typical of two-dimensional turbulence in the presence of spectral condensation [100] or 3*D* rotating turbulence at moderate Rossby number [101]. Dominance of the large-scale 2*D*-mode is evidenced by a higher energy content at smaller wave-numbers compared to that of the 3*D*-mode, while flattening of the 3*D* mode at small κ indicating presence of large-scale overturning structures [99]. In general, the spectra of both modes upscale from their local minima at $\kappa \lesssim 100$ are similar to those found by the NH-QGS model for free-slip boundaries [47] at $\widetilde{\mathrm{Ra}} = 90$, $\mathrm{E} = 10^{-7}$, and by the DNS and the Ekman pumping-supplemented NH-QGS model in [41]. The inertial range for the 3*D* mode is somewhat larger in Fig. 19 than that reported in [41,47] which might be due to the higher Rayleigh number in the present simulation.

However, present simulation also demonstrates a strong and relatively narrow peak downscale in both spectra. Similarly shaped peaks were also observed in the simulation of forced 2*D* turbulence [100] and of forced rotating stratified turbulence [102], at the corresponding scales associated with the forcing. We speculate that apparent forcing in the present high-Rayleigh number simulation is an effect of vortical dynamics in the middle sub-layer and its role is to enhance the otherwise non-local inverse cascade [40,41,47] of the 2*D*-mode to produce a sustainable large-scale circulation. Certainly, this question needs a much deeper study with more resolution for the Ekman layer, but it goes beyond the purpose of the present contribution. To summarize, simulation of rapidly rotating RBC hints at the appearance of sustainable large-scale circulation also in the case of no-slip boundaries, once the Rayleigh number is large enough.

6. Conclusion

In this paper, we have accessed capabilities of the new two-population entropic multi-relaxation lattice Boltzmann in a range of problems related to turbulent thermal convection. Extensive simulation plan was carried out for both non-rotating and rotating Rayleigh-Bénard convection, including the demanding rotationally constrained situation for the latter. Select cases were compared to existing DNS and asymptotic reduced models. Overall, this study demonstrates that even with the modest grid resolution requirements as compared to both DNS and LES, the present KBC model is capable of quantitatively capturing the low-order statistics of the flow and temperature dynamics. In our study, we used the uniform grid in order to present a compelling, simplest lattice Boltzmann algorithm. However, enhanced capabilities are readily available when using grid refinement strategies [87]. Also, more isotropic lattices with enhanced Galilean invariance can be used in order to increase the range of velocity magnitude in the low Mach number lattice Boltzmann formulation [103]. It is thus possible that the present model can serve a viable alternative to more conventional LES, RANS and other methods to study demanding regimes of thermal convection.

Acknowledgments

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